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AI for Linux on IBM Z and LinuxONE Applications and Examples

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Contents

Notices v
Trademarks vi

Preface vii
Authors vii
Now you can become a published author, too! ix
Comments welcome ix
Stay connected to IBM Redbooks x

Chapter 1. Introducing the convergence of artificial intelligence and enterprise Linux systems 1
1.1 Entering the era of the multiple-AI model architecture 2
1.2 Why use AI on IBM Z and LinuxONE 4
1.3 IBM Z and LinuxONE ecosystem overview 5
1.4 Integrated Accelerator for AI 6
1.5 Migrating an AI application to IBM Z and LinuxONE 6
1.5.1 Migrating an AI application to Red Hat OpenShift, IBM Z, and LinuxONE 7

Chapter 2. Optimizing model serving solutions 9
2.1 AI Toolkit for IBM Z and LinuxONE 10
2.1.1 IBM Z Accelerated for TensorFlow 12
2.1.2 IBM Z Accelerated for TensorFlow Serving 13
2.1.3 IBM Z Accelerated for Snap ML 14
2.1.4 IBM Z Deep Learning Compiler 15
2.1.5 IBM Z Accelerated for NVIDIA Triton Inference Server 15
2.1.6 IBM Z Accelerated for PyTorch 17
2.2 IBM Cloud Pak for Data 19
2.2.1 Overview of services that are available 20
2.2.2 Getting started with Db2 Data Gate 20
2.2.3 AI governance 21

Chapter 3. Building enterprise-ready artificial intelligence applications at scale . . . 23
3.1 Reasons to deploy AI solutions on Linux on IBM Z and LinuxONE 24
3.1.1 Reliable and available 25
3.1.2 Secure 25
3.1.3 Scalable 26
3.1.4 AI acceleration 26
3.1.5 Available AI frameworks and tools for Linux 27
3.2 The AI lifecycle methodology 28
3.2.1 Starting the Proof of Concept: Personas 28
3.2.2 Project planning for a PoC and production implementation 29
3.2.3 Example implementation of a PoC 30
3.3 Examples of use cases 31
3.3.1 Predictive AI 33
3.3.2 Generative AI 34
3.3.3 AI multiple-model architecture 34
3.3.4 AI Security 35

Chapter 4. Bringing it all together with the use cases 37

4.1 Use case: Advanced multi-model AI framework for home insurance fraud detection processing 38
4.1.1 Overview, challenges, and needs for fraud detection in insurance 38
4.1.2 Advanced multi-model AI insurance claims fraud detection: A reference architecture on IBM Z and LinuxONE 39
4.2 Use case: An advanced multi-model AI framework for anti-money laundering 40
4.2.1 AML implementation: Overview, challenges, and requirements 41
4.2.2 Reference architecture of the Multi-Model AI AML framework on LinuxONE 41
4.3 Use case: Credit risk assessment 44
4.4 Overview of additional use cases 48
4.5 Additional solution templates and resources to get started with AI on Linux 51

Abbreviations and acronyms 53

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Preface

You can accelerate your enterprise without compromising on security or flexibility. Use artificial intelligence (AI) for Linux on IBM Z® and LinuxONE to jump start your AI projects on a robust platform for developing and deploying AI applications.

This IBM Redbooks® publication is a practical guide with real examples and use cases to help you use the efficiency and security of Linux on IBM Z and LinuxONE to accelerate your AI journey. It also helps you understand the entry points and steps that are involved in deploying an AI application.

This IBM Redbooks publication shows you how AI can solve complex problems and drive innovation on some of the most powerful hardware available and includes the following topics:

- Chapter 1, "Introducing the convergence of artificial intelligence and enterprise Linux systems" on page 1
- Chapter 2, "Optimizing model serving solutions" on page 9
- Chapter 3, "Building enterprise-ready artificial intelligence applications at scale" on page 23
- Chapter 4, "Bringing it all together with the use cases" on page 37

Whether you are a developer, a business owner, or simply curious about the future, this IBM Redbooks publication helps you understand how implementing AI in your enterprise can solve complex problems and drive innovation on powerful hardware.

Throughout this publication, Linux on IBM Z and LinuxONE are referred to as Linux because they both run on the s390x architecture.

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He is a research scholar pursuing a doctorate in AI, bringing both academic insight and hands-on leadership to his work. He has led numerous cloud development projects and has been deeply involved in IBM Z modernization initiatives. He collaborates closely with global system integrators, managed service providers and the Global Capability Centre to modernize mainframe application and integrate next-generation capabilities.

As a recognized IBM Z Advocate, Abraham actively promotes IBM Z technologies, mentors emerging talent, and contributes to the growth of the global IBM Z community. He has authored several technical blogs and co-authored white papers with system integrators.

Markus Wolff has been a technical specialist for IBM Z solutions, focusing on Data and AI on IBM Z since 2019. He holds a bachelor's degree in business administration with focus on Industry 4.0 and digitization. During his studies, he focused on digitizing business models and the economic aspects of IT operations analytics and machine learning (ML).

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He has contributed to IBM Redpapers regarding AI on IBM Z, such as *Optimized Inferencing and Integration with AI on IBM zSystems: Introduction, Methodology, and Use Cases*, REDP-5661, and is part of the worldwide AI on IBM Z SWAT team, led by IBM Fellow Elpida Tzortzatos.

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1

Introducing the convergence of artificial intelligence and enterprise Linux systems

Artificial intelligence (AI) is a technology that simulates human intelligence by learning from data, making decisions, and improving over time.

AI systems process large amounts of data, identify trends, and make predictions. You can use AI to perform tasks that typically require human intelligence, such as speech recognition, visual perception, and language translation. AI includes a wide range of technologies, including natural language processing (NLP), machine learning (ML), robotics, and computer vision.

You can use AI to mimic human intelligence and perform tasks that require human cognition by using a combination of data, algorithms, and computational power.

Here is a basic summary of how AI works:

- Data gathering: You provide AI with large amounts of data from sources such as text, numbers, and images.
- Learning: AI uses algorithms, including ML and deep learning (DL), to find patterns, relationships, or rules in the data.
- Processing and decision making: AI creates a model and uses learned patterns to make predictions or suggestions.
- Enhancement: You can improve AI by enabling it to refine decisions by using new data and self-learning.

1.1 Entering the era of the multiple-AI model architecture

AI evolved over decades, shifting from theoretical concepts to a transformative force across industries. In the 1950s, AI emerged with foundational ideas like Alan Turing's work on machine intelligence and the development of early rule-based systems. In the 1980s and 1990s, researchers advanced ML with more powerful computers and neural network algorithms, although progress was limited by data and hardware constraints. In the 21st century, the rise of big data, cloud computing, and DL enabled breakthroughs in NLP, computer vision, and generative models. You can view this evolution in the timeline shown in Figure 1-1.

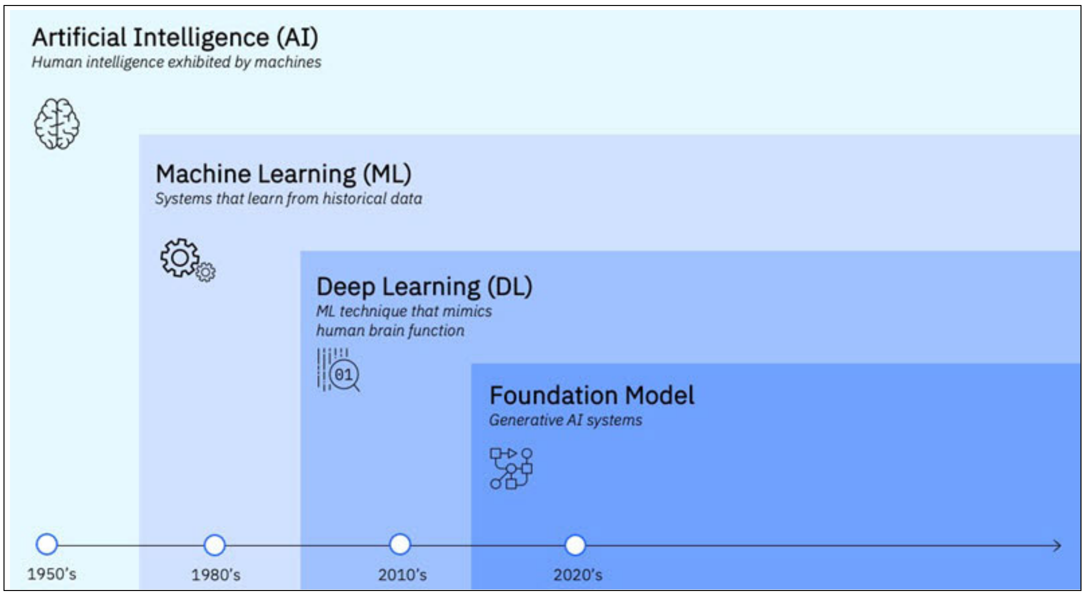

Figure 1-1 An evolution of AI technologies over time

Today, AI includes multi-model AI architectures that support both generative and predictive AI capabilities. These architectures are designed to handle diverse tasks by processing data types such as text, images, and audio. You can find more details and a sample use case in 3.3.3, "AI multiple-model architecture" on page 34.

The AI landscape includes several key families of algorithms, each contributing to its diverse capabilities. You can explore the following examples:

- Machine learning (ML)

  You provide inputs, and the ML-based systems analyze historical data to identify patterns and generate corresponding outputs. ML is commonly used for predictive and classification tasks, such as forecasting house prices or detecting fraudulent business transactions.

  You can apply ML methods including supervised learning, semi-supervised learning, unsupervised learning, and reinforcement learning.

- Deep learning (DL)

  DL is a subset of ML that uses multiple layers of neural networks. These networks consist of interconnected nodes that work together to process information. DL is suitable to complex applications such as image and speech recognition.

- Generative AI

  Generative AI models use a neural network architecture that is called a transformer, which generates sequences of related data elements such as sentences. You can use generative AI to create new content, including text, audio, images, or videos.

The introduction of foundation models, also known as large language models (LLMs), marked a significant breakthrough for generative AI. LLMs are pre-trained on extensive cross-industry datasets and can contain millions or even billions of parameters, reflecting their scale. Generative AI supports a wide range of applications, such as translation and image generation. However, this publication focuses on business use cases.

LLMs introduce new challenges for AI infrastructure. You need substantial compute resources to deploy, train, fine-tune, and govern these models. Governance is essential because LLMs can hallucinate over time, generating grammatically correct content that lacks meaningful substance.

You can gain deeper insights, improve accuracy, and build more robust and effective business systems by combining predictive and generative AI technologies in business applications. You can reduce response times by deploying multiple AI models alongside transactional systems. This approach is especially important for generative AI use cases, which might respond slower because of the architectural complexity of LLMs.

IBM z17® and LinuxONE 5 platforms provide the infrastructure that you need to implement multiple AI model use cases. These platforms offer a sustainable environment designed for AI and generative AI workloads that are critical to your business. Hardware acceleration, through the IBM Telum® II co-processor or IBM Spyre™ AI accelerator makes converged AI scalable and resilient. It dynamically adjusts computing power to meet your service-level agreements (SLAs).

You can use Table 1-1 to compare key features of Telum and Telum II.

Table 1-1 Telum and Telum II comparison chart

Feature	Telum	Telum II
On-chip AI acceleration	IBM z16® A01	IBM z17 ME1[a]
Support for LLM compute primitives	No	Yes
Technology node	7 nm	5 nm
Processor cores	8 cores	8 cores
Capacity	200 characterizable processor units	208 characterizable processor units
Clock Speed	Over 5 GHz	5.5 GHZ
Cache	32 MB L2 per core, 256 MB virtual L3, and 2 GB virtual L4	36 MB L2 per core, 360 MB virtual L3, and 2.88 GB virtual L4
AI Acceleration	On-chip AI accelerator for inferencing	Designed for enhanced AI accelerator with 4x compute power
Data processing unit (DPU)	Not available	Integrated DPU for I/O acceleration

Feature	Telum	Telum II
Security	Transparent encryption of main memory	Designed for enhanced security features, including quantum-safe methods
Performance	Optimized for real-time analytics and decision-making	Designed for improved performance for AI-driven workloads and enterprise transactions

a. The calculations that are presented in this column were established through internal measurements. Your results might vary based on individual workload, configuration, and software levels. For more information, see the LSPR web page or *IBM z17 Technical Introduction*, SG24-8580.

Also, you have access to a rich software ecosystem that includes frameworks for end-to-end AI model development, training, inferencing, and governance. You can train models off-platform and bring them back in open standard formats, such as Open Neural Network Exchange (ONNX) and Predictive Model Markup Language (PMML), for high-performance inferencing on the IBM Z platform.

1.2 Why use AI on IBM Z and LinuxONE

AI demands significant computational power and relies on specialized architecture, making a strong and scalable infrastructure essential to unlocking its full potential. Without this foundation, you cannot develop and deploy state-of-the-art AI solutions. Infrastructure plays a critical role throughout the AI lifecycle. Platforms such as IBM Z and LinuxONE support mission-critical workloads and sensitive data. However, using AI in high-performance environments presents unique challenges, such as delivering real-time insights while maintaining strict SLAs and protecting data privacy.

A key emerging trend is the growing adoption of placing AI models alongside critical business applications on IBM Z and LinuxONE. This strategy enables you to make faster, more informed decisions by processing data directly on-platform, without needing to move data elsewhere. By keeping AI processing close to where the data resides, you enhance security, reduce latency, and accelerate the delivery of impactful outcomes.

If you want to use AI in real-time transactions involving large volumes of data, Linux on IBM Z and LinuxONE platforms offer several advantages:

- High performance: These platforms include integrated AI accelerators, such as the IBM Telum processor, which significantly boosts AI inferencing performance. You can process large volumes of transactions with low latency.
- Scalability: IBM Z and LinuxONE are designed to handle massive amounts of data, making them ideal if you need to scale your AI applications. They manage data across platforms efficiently, helping ensure seamless integration and high throughput.
- Security and compliance: The AI Toolkit for IBM Z and LinuxONE includes IBM Elite Support and IBM Secure Engineering. These tools help ensure that AI frameworks remain secure and comply with industry regulations.

- Cost-effective: These platforms offer cost-effective entry points for using AI. They reduce costs and complexity while accelerating time to market with lightweight tools and runtime packages.
- Power efficiency: These platforms enhance power efficiency by consolidating Linux workloads, which reduces energy consumption. Also, these systems can help reduce CO2 emissions, supporting more sustainable IT operations.

Chapter 2, “Optimizing model serving solutions” on page 9 highlights two key products that empower AI innovation. The AI Toolkit for IBM Z and LinuxONE helps you use popular open-source AI frameworks, such as TensorFlow and PyTorch, directly on the platform. IBM Cloud Pak® for Data runs on both Linux on IBM Z and LinuxONE and provides a cloud-native platform for managing AI and analytics. It enables you to efficiently build, deploy, and govern AI models directly where the data resides. The overall goal of AI on these platforms is to help you make faster, smarter decisions without the complexity of moving data.

AI makes databases smarter and more efficient. IBM Db2 AI for z/OS uses ML to enhance query performance by optimizing how data is accessed and processed. With AI-driven tools such as IBM watsonx Assistant™ and IBM watsonx Code Assistant®, you can automate tasks, simplify interactions, and accelerate application modernization on IBM Z and LinuxONE.

IBM Z and LinuxONE provide a dedicated software ecosystem and full-stack hardware architecture, creating a comprehensive platform for AI acceleration, as described in 1.4, “Integrated Accelerator for AI” on page 6. The Telum and Telum II processors feature an on-chip AI accelerator that enables high-speed AI inferencing. For more demanding workloads, the Spyre accelerator provides more AI processing power. You can use IBM Z and LinuxONE to handle billions of AI inference requests daily, enabling real-time AI at enterprise scale.

1.3 IBM Z and LinuxONE ecosystem overview

IBM Z and LinuxONE offer operating system and virtualization options that include IBM z/OS, IBM z/OS Container Extensions (zCX), Ubuntu, SUSE, and Red Hat OpenShift. You can run containerized workloads flexibly, integrate with cloud-native environments, and support modern DevOps practices.

This section provides an overview of the ecosystem, including a wide range of programming languages, libraries, and compilers such as Python, Java, Go, GCC, and C++. You can use optimized libraries, such as OpenBLAS and DL Compiler, to accelerate AI computations and support efficient workload execution.

For IT operations, the platform includes IBM Db2 AI for z/OS, IBM Z Anomaly Analytics, IBM Cloud Pak for AIOps, and IBM Instana® for real-time monitoring and observability. You can use these tools to optimize system performance, detect anomalies, and improve operational efficiency.

IBM Z and LinuxONE provide key AI and analytics capabilities, which include IBM Machine Learning for z/OS, the Python AI Toolkit, IBM Synthetic Data Sets, Cloud Pak for Data, the AI Toolkit for IBM Z and LinuxONE, and IBM Db2 with SQL Data Insights. You can use these tools to apply AI and ML for data-driven decisions. Db2 Analytics Accelerator and IBM Watson® further enhance your ability to analyze large volumes of structured and unstructured data efficiently.

IBM Z and LinuxONE support various AI and data processing frameworks, including PyTorch, TensorFlow, Apache Spark, Scikit-learn, and Anaconda. These frameworks support advanced ML, DL, and big data analytics, making it simpler for you to deploy AI workloads on the platform.

Organizations across industries, including banking, finance, insurance, retail, hospitality, healthcare, and government use IBM Z and LinuxONE. These sectors benefit from the platform's enterprise-grade security, high availability, and scalability.

1.4 Integrated Accelerator for AI

At the heart of IBM z17 is the Telum II processor and the new on-chip Integrated Accelerator for AI. The integrated AI accelerator uses dedicated on-chip silicon to deliver high performance and consistently low-latency inferencing for processing a mix of traditional transactional and AI workloads at speed and scale.

With the on-chip AI accelerator, you can integrate complex AI inferencing with real-time data into transactional workloads and still meet even the most stringent SLAs. This integration enables you to deliver insights for every transaction without compromising performance.

The IBM z17 and IBM LinuxONE Emperor 5 are designed to process up to 5 million inference operations per second, or approximately 450 billion inference operations per day, with less than a 1-millisecond response time by using a credit card fraud detection DL model.

1.5 Migrating an AI application to IBM Z and LinuxONE

Some reasons to migrate your AI applications to Linux include the following ones:

- IBM Z and LinuxONE offer a highly available, resilient environment with built-in fault tolerance, helping you run AI applications with minimal downtime.
- You can consolidate workloads to improve cost efficiency by reducing the need for multiple physical or virtual servers.
- You can integrate AI into core business processes seamlessly, with lower latency and enhanced security.

IBM Z and LinuxONE give you the flexibility to deploy AI across hybrid cloud environments. You can train and run AI workloads on-premises or in the cloud with providers such as AWS, Azure, and IBM Cloud®, making AI deployment frictionless and scalable.

Figure 1-2 on page 7 provides a migration example. On the left side of the figure, an AI application interacts with TensorFlow, running on Linux, hosted on a distributed system. This setup relies on multiple servers or a cloud-based infrastructure, which can create challenges in security, scalability, and operational management. AI workloads require substantial computing power, which can result in high operational costs and added complexity when managing multiple nodes.

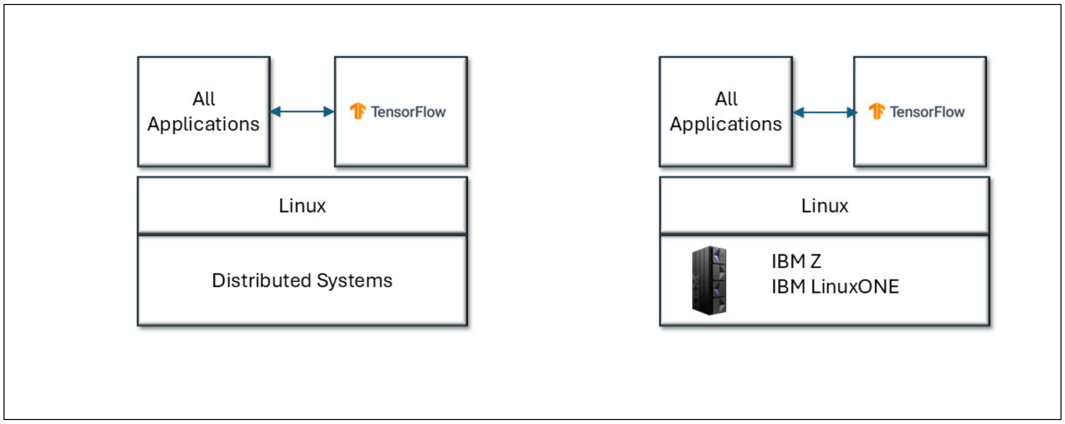

Figure 1-2 Migrating an AI application to IBM Z and LinuxONE

On the right side of the figure, the AI application runs on IBM Z and LinuxONE. The AI application and TensorFlow continue to run on Linux, but instead of being hosted on a distributed system, they now operate on Linux on IBM Z or LinuxONE. This shift provides key advantages, including enhanced security features such as pervasive encryption and Secure Execution, which help ensure that your AI workloads remain protected. The AI accelerator capabilities in IBM Z improve efficiency and performance, making AI inference and model training faster and more reliable.

1.5.1 Migrating an AI application to Red Hat OpenShift, IBM Z, and LinuxONE

As shown in Figure 1-3, when you migrate AI applications to Red Hat OpenShift on IBM Z and LinuxONE, you can achieve a more scalable, secure, and efficient deployment model. Scaling Linux environments on distributed systems often requires provisioning multiple servers. This approach increases complexity in workload management, adds latency, and reduces overall reliability.

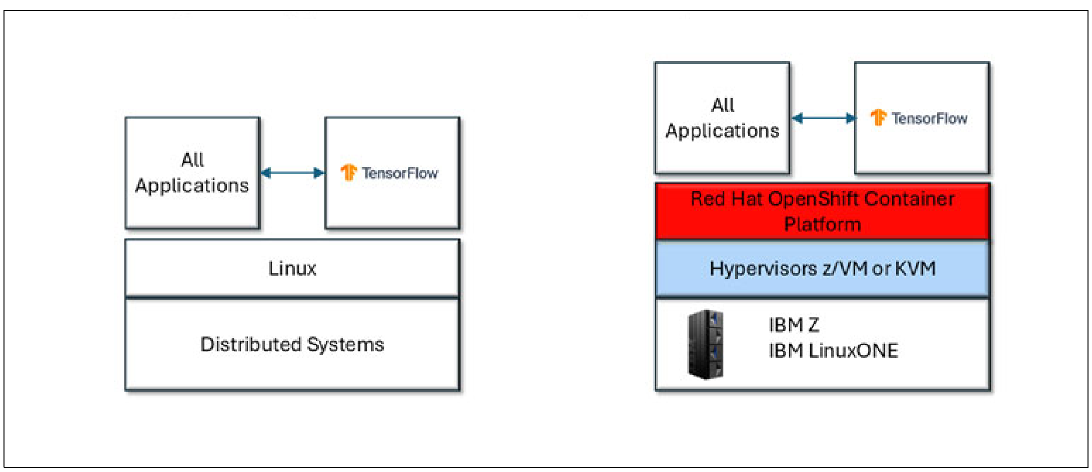

Figure 1-3 Migrating an AI application to IBM Z and LinuxONE with Red Hat OpenShift

When you migrate an application architecture to IBM Z and LinuxONE with Red Hat OpenShift, the AI application and TensorFlow no longer run directly on Linux. Instead, they run in containers that are managed by the Red Hat OpenShift Container Platform. This setup uses hypervisors, such as z/VM or KVM, to manage virtualization and optimize resource allocation. By using OpenShift AI, you can take advantage of container orchestration, automated scaling, and improved deployment across environments.

By moving AI workloads to IBM Z and LinuxONE with Red Hat OpenShift:

- You gain a highly available, resilient environment with built-in fault tolerance, helping ensure AI applications run with minimal downtime.
- You can simplify infrastructure management and enhance security because IBM Z and LinuxONE provide industry-leading encryption and isolation capabilities.
- Running AI workloads in containers helps ensure better portability and agility, making it simpler to integrate with modern DevOps work flows.
- With Red Hat OpenShift, you can scale AI applications dynamically based on demand and reduce the impact of managing individual virtual machines or physical servers.

IBM Z and LinuxONE enable seamless AI integration into enterprise workloads across industries such as banking, finance, healthcare, retail, and government. You can build and train AI models anywhere by using frameworks such as Python, Jupyter, TensorFlow, PyTorch, Keras, and Snap ML. You can then optimize these models by using ONNX and deploy them efficiently on IBM Z hardware through IBM Snap ML or the DL Compiler by using operating systems such as z/OS, Linux, and Red Hat OpenShift.

2

Optimizing model serving solutions

This chapter describes the following solutions for optimized model serving:

- AI Toolkit for IBM Z and LinuxONE
- IBM Cloud Pak for Data

2.1 AI Toolkit for IBM Z and LinuxONE

You can access the AI Toolkit for IBM Z and LinuxONE, which includes a collection of machine learning (ML) and deep learning (DL) serving solutions, from the IBM Z and LinuxONE Container Registry. IBM Elite Support is included. You can use the toolkit as a complete bundle or choose individual components based on your needs.

The AI Toolkit for IBM Z and LinuxONE includes six high-performance ML and DL frameworks: TensorFlow, TensorFlow Serving, Snap ML, Triton, PyTorch, and the DL Compiler.

> **Note:** For more information about the AI Toolkit for IBM Z and LinuxONE, see AI Toolkit for IBM Z and LinuxONE.

These frameworks and serving solutions are optimized to use the Telum on-chip artificial intelligence (AI) accelerator, which delivers efficient inference and scoring of traditional AI workloads in both ML and DL.

The AI Toolkit for IBM Z and LinuxONE includes enterprise-grade IBM Elite Support to help you deploy open-source into production on enterprise systems.

In addition to support, IBM applies a secure engineering process to these open-source AI libraries. IBM scans, vets, and continuously monitors them for vulnerabilities.

For more information, see 3.1.4, “AI acceleration” on page 26.

Figure 2-1 provides an overview of the high-level architecture of the AI Toolkit for IBM Z and LinuxONE.

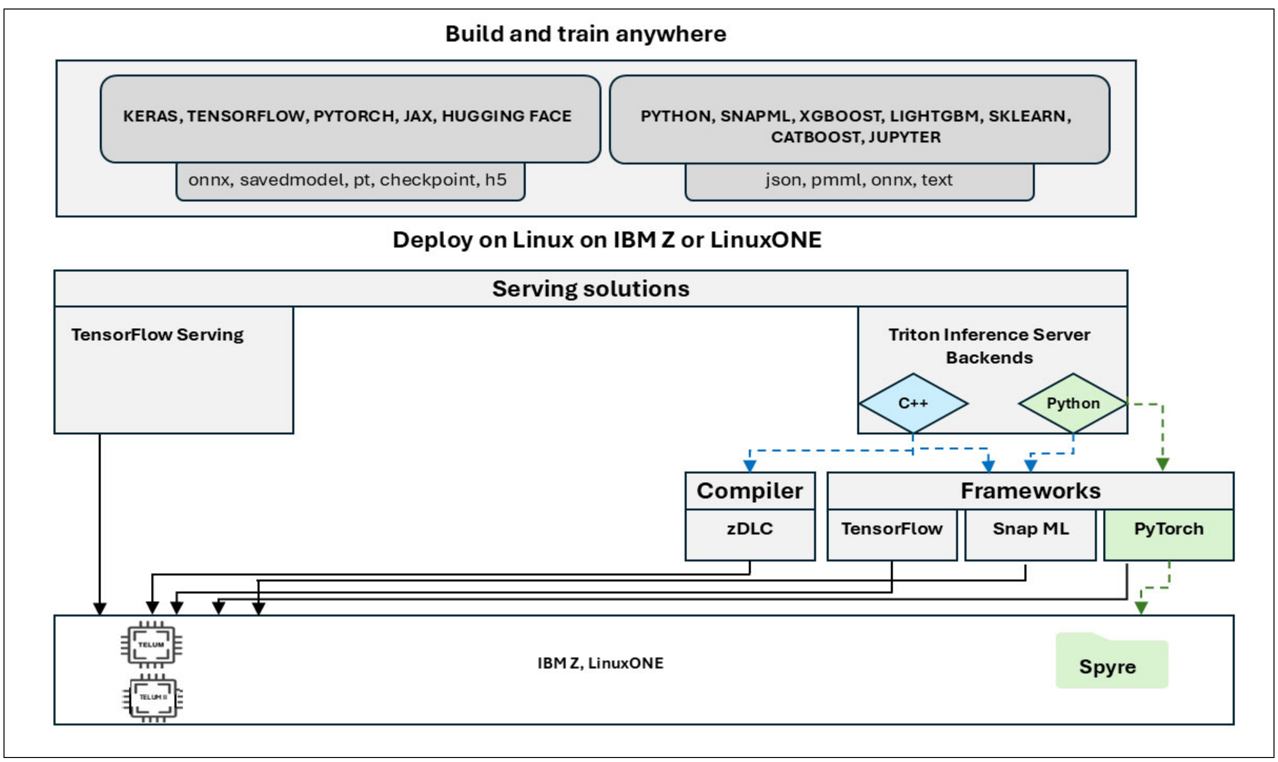

Figure 2-1 AI Toolkit for IBM Z and LinuxONE high-level architecture

The AI Toolkit for IBM Z and LinuxONE architecture provides flexibility for model development and efficient deployment on IBM hardware, including IBM Z and LinuxONE with on-chip accelerators.

Figure 2-1 on page 10 illustrates the following layers:

1. The Build and Train Anywhere layer
   - This layer supports mainstream libraries and frameworks, including:
     - DL: Keras, TensorFlow, PyTorch, JAX, and Hugging Face for DL.
     - Traditional ML and experimentation: Python, Snap ML, XGBoost, LightGBM, scikit-learn, CatBoost, and Jupyter.
   - You can export model artifacts in standard formats such as Open Neural Network Exchange (ONNX), SavedModel, PyTorch (.pt), Checkpoint, HDF5 (.h5), and structured data formats, like JSON, Predictive Model Markup Language (PMML), and plain text.
   - You can deploy models by using serving solutions available on the IBM Z platform:
     - TensorFlow Serving: Optimized for TensorFlow models.
     - Triton Inference Server: Supports C++ and Python back ends extending compatibility across frameworks.
2. Deploy: Serving options for optimized AI frameworks:
   - TensorFlow: A widely used DL framework.
   - Snap ML: IBM's high-performance ML library.
   - IBM Z Deep Learning Compiler (zDLC).
   - PyTorch: Integrated through a Python back end that is served by using Triton Inference Server.

   You can deploy the serving stack on IBM z16 or z17 systems, which use the Telum chip to provide on-chip AI inferencing.

   These frameworks are tuned to take advantage of this hardware, enabling low latency, and high-throughput AI inference.

In the lower right of Figure 2-1 on page 10, Spyre appears as an application integration layer that enables business applications and logic to interact natively with AI models. It bridges the gap between data, production systems, and models.

This architecture supports the complete AI lifecycle from training with your preferred toolset to optimized deployment on Linux. It combines the flexibility of open source with high-performance serving and enterprise-grade reliability, which is powered by IBM hardware and AI acceleration.

These frameworks form the AI Toolkit for IBM Z and LinuxONE.The deployment options appear in Figure 2-2 and are discussed in the following sections.

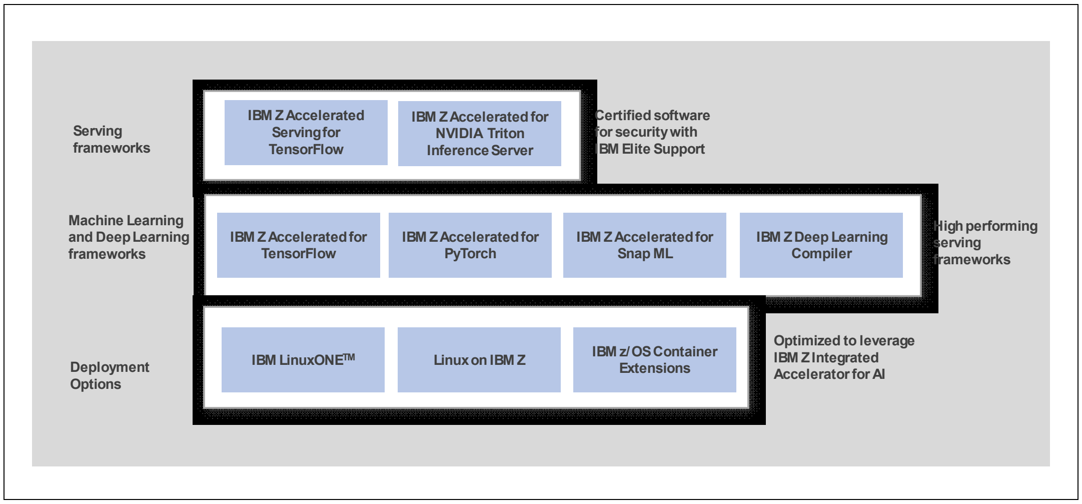

Figure 2-2 Frameworks and deployment options in AI Toolkit for IBM Z and LinuxONE

Note: All serving, ML, and DL frameworks are accelerated for both IBM Z and LinuxONE.

2.1.1 IBM Z Accelerated for TensorFlow

TensorFlow provides open-source DL tools and a flexible framework for data scientists. You can use TensorFlow to apply ML and DL libraries across a wide range of use cases, including natural language processing (NLP) and text-based applications, image recognition, voice search, and more. TensorFlow powers systems such as Facebook's DeepFace image recognition platform and Apple's Siri voice recognition service. It offers a scalable, flexible system for defining, training, and deploying ML models by managing data flow and computations with robust support for DL and hardware optimization.

TensorFlow enables you to integrate DL & ML into mission-critical workloads at scale by using Integrated Accelerator for AI.

You can access TensorFlow through the following methods:

1. Linux distributions: Available on IBM z/OS Container Extensions (zCX), Linux on IBM Z and LinuxONE, and through distribution channels such as Docker Hub
2. IBM Cloud Pak for Data: Supported in Version 4.6 and later
3. IBM LinuxONE container image repository

Options 2 and 3 are designed to provide optimal performance and validated security.

You can view the TensorFlow-powered model in Figure 2-3 on page 13:

1. Receives input data from different classes (clusters), shown in Figure 2-3 on page 13 as ω1, ω2, and ω3.
2. Learns to distinguish between classes through training.

3. Generates predictions or inferences based on input data by using the following approaches:
   - Classification: Produces probabilities or class labels, such as [0.9, 0.1] for a binary classifier or [“cat”] for image recognition.
   - Regression: Produces numerical values, such as 42.5 for a house price prediction.
   - Generative Models: Produce data such as images, text, or audio, such as an image tensor generated from a Generative Adversarial Network (GAN).

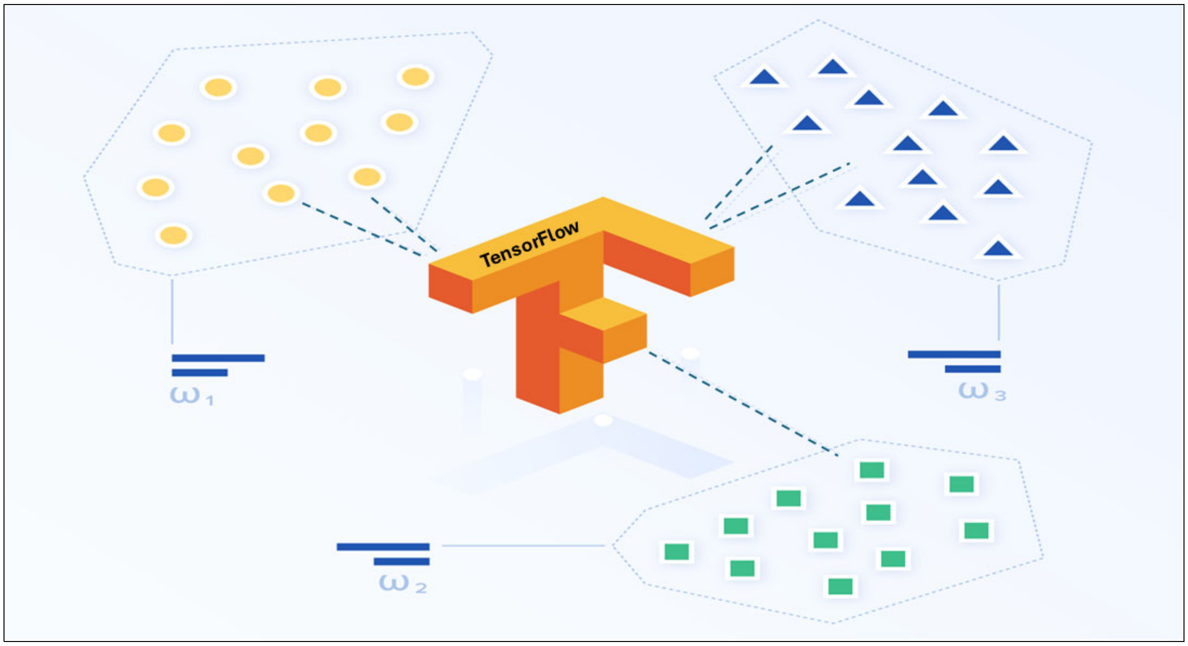

Figure 2-3 TensorFlow with Linux

2.1.2 IBM Z Accelerated for TensorFlow Serving

This section describes TensorFlow Serving and explains how to deploy a TensorFlow model for inferencing. TensorFlow Serving is a library for serving ML models developed with TensorFlow. It offers a rich set of features, including model versions, automatic request batching, server-side batching, and client-side batching.

TensorFlow Serving delivers predictions through HTTP REST or Google Remote Procedure Call (gRPC) interfaces. These interfaces support high-throughput, low-latency inference serving.

Figure 2-4 presents an ML model deployment that uses TensorFlow Serving.

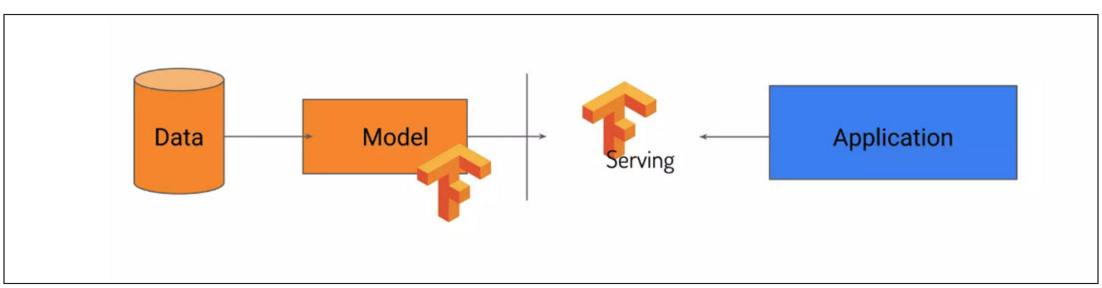

Figure 2-4 Machine learning model deployment by using TensorFlow Serving

The following steps provide a high-level overview of ML model that uses TensorFlow Serving:

1. Train your model on data.
2. Export the model by using TensorFlow.
3. Deploy the model with TensorFlow Serving.
4. Integrate the model into a real-world application for inference.

TensorFlow uses static computation graphs to support production scalability, deployment efficiency, and operational performance. Although PyTorch offers flexible debugging and ease of use, as discussed in 2.1.6, “IBM Z Accelerated for PyTorch” on page 17), TensorFlow remains more widely adopted in business environments because of its mature tools.These tools include TensorFlow Serving and TensorFlow Lite, which support enterprise and mobile applications.

TensorFlow integrates with IBM platforms such as Watson Machine Learning and Cloud Pak for Data. These solutions enable you to build and deploy AI models for use cases such as predictive maintenance, medical image analysis, chatbot development, and perform vision-based tasks in retail and security. Most businesses use TensorFlow for predictive modeling and other DL tasks. It supports real-world applications by delivering tailored solutions to complex challenges faced by large-scale enterprises.

2.1.3 IBM Z Accelerated for Snap ML

Snap ML is an open-source library that is developed and maintained by IBM Research® that accelerates the training and inference of ML models.

It provides high-performance implementations for the following models:

- Generalized linear models
- Tree-based models
- Gradient boosting models

Snap ML supports the IBM Z Integrated AI Accelerator in Telum and Telum II. Snap ML is included in the AI Toolkit for IBM Z and LinuxONE, IBM Cloud Pak for Data (which runs on both Linux on IBM Z and LinuxONE), and IBM Machine Learning for IBM z/OS (MLz). It is also available as a stand-alone container image through PyPi.

For more information, see this GitHub repository.

You can find example notebooks that demonstrate how to use the IBM Snap ML library at this GitHub repository.

Snap ML supports both linear and tree-based ML algorithms. Figure 2-5 on page 15 shows several of the learning algorithms.

Figure 2-5 Linear and tree-based machine learning algorithms

2.1.4 IBM Z Deep Learning Compiler

zDLC enables you to build and train DL models by using your choice of open-source frameworks, such as TensorFlow or PyTorch. It supports C++, Java, and Python APIs.

You compile DL models optimized to use the Integrated Accelerator for AI and deploy them on IBM LinuxONE 4, IBM z16, and IBM z17.

The compiled shared library files can be embedded in applications independent of the original framework's dependencies and packages.

For more information about zDLC, see this GitHub repository.

2.1.5 IBM Z Accelerated for NVIDIA Triton Inference Server

The NVIDIA Triton Inference Server is open-source software that supports AI model deployment and execution. It is a high-performance model inference framework optimized by the AI on IBM Z team for IBM Z and LinuxONE architectures.

Triton offers the flexibility to create custom model back ends. As part of the AI Toolkit for IBM Z and LinuxONE, the focus is on the following back ends:

- Traditional ML models in the PMML, ONNX, or JSON format run by using the IBM Snap ML run time.
- DL models in ONNX model format, which are compiled with the zDLC.

Key Features of the Triton Inference Server include:

- High-performance inference serving ML and DL models at scale
- Multiple framework support
- Model ensemble support
- Concurrent model execution with dynamic batching
- Service-level agreement (SLA)-aware performance optimization
- Availability as a Triton container image from the IBM Cloud Container Registry

The Triton Inference Server is designed to run customer transaction predictions up to 3.5x faster by colocating the application with the Snap ML library on IBM LinuxONE 4, compared to running predictions remotely by using the NVIDIA Forest Inference Library on a comparable x86 server.

Figure 2-6 shows the performance difference between running an application colocated with the Snap ML library on IBM LinuxONE and running it remotely.

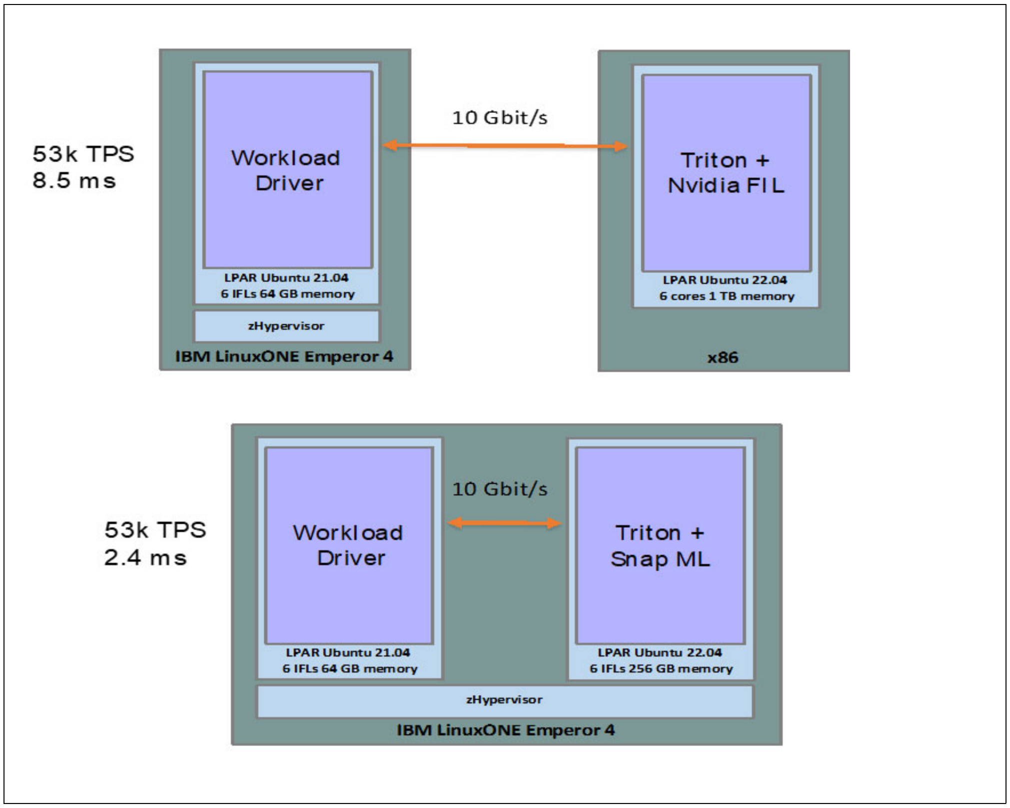

Figure 2-6 Comparing AI inference deployment: IBM LinuxONE with Snap ML versus x86 with NVIDIA Forest Inference Library

Combining data pre-processing and inference by using low-level programming languages reduces overall inference time and is essential for meeting strict SLAs.

2.1.6 IBM Z Accelerated for PyTorch

PyTorch supports Linux environments, including the s390x architecture used by IBM LinuxONE. Official PyTorch releases are available through package managers, such as pip and conda, which include builds for s390x. IBM actively contributes to maintaining compatibility, particularly for enterprise workloads.

PyTorch-based AI applications benefit from IBM Z and LinuxONE features, including encrypting data in transit, at rest, and in use through confidential computing.

PyTorch is a fully featured framework for building DL models and is designed to provide flexibility and high performance for deep neural network implementations.

PyTorch on Linux for IBM Z enables you to integrate DL and ML into mission-critical workloads at scale by using Integrated Accelerator for AI. It is available from the IBM LinuxONE container image repository.

Here are the key benefits of PyTorch with Linux on IBM Z:

- Supports both traditional AI and foundation model stacks on IBM Z.
- Enables deployment of transformer and encoder-based models, and ensemble AI models.
- Optimized for IBM Telum, IBM Telum II, and Spyre.
- Allows you to import pre-trained models and use AI accelerators for faster deployment.
- Offers a dynamic computation graph that provides an intuitive framework for development and experimentation.

PyTorch is especially suitable for AI-based research and experimentation, including natural language understanding with Hugging Face Transformers. Evolving fields such as medical AI, federated learning, and generative models (including AI-based image and text generation) benefit from PyTorch's ability to support rapid prototyping of new concepts.

Figure 2-7 shows an architecture that seamlessly integrates AI and ML capabilities into enterprise-class transactional systems while maximizing reuse of existing IBM infrastructure. This architecture is optimized for performance, flexibility, and enterprise scalability, using open-source frameworks and IBM-optimized toolkits.

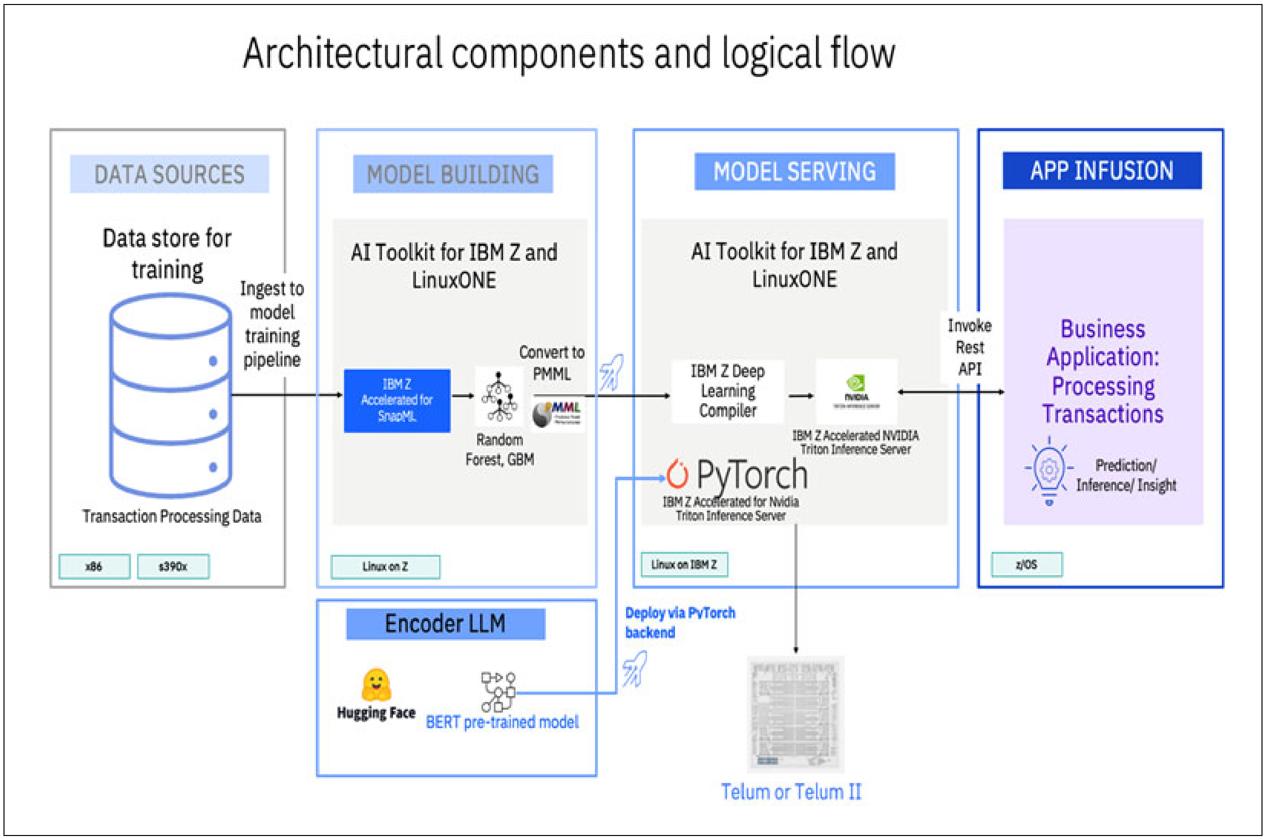

Figure 2-7 End-to-end architecture for AI integration in business applications with Linux

The data flow includes data ingestion, model training and deployment, model serving, and application integration.

The following list describes each section that is shown in Figure 2-7:

- Data sources

  In this example, a training data store containing transaction processing data that originates from x86 or s390x platforms. This data feeds into the model training pipeline to build AI models.

- Model building

  During the model building phase, you use the AI Toolkit for IBM Z and LinuxONE. IBM Z optimized Snap ML algorithms, such as Random Forest and Gradient Boosting Machines (GBM), are trained and exported to PMML for standardization and portability.

  Encoder-based large language models (LLMs), such as pre-trained BERT models from Hugging Face, are also supported and are useful for NLP tasks.

  These models are served through the PyTorch backend on IBM Z infrastructure.

  AI quantization can be applied during or after model training.

  This method reduces numerical precision, for example, from 32-bit floats to 8-bit integers, without compromising the integrity of the model's output.

The benefits include the following items:

– Increased computational speed
– Reduced memory footprint
– Reduced energy consumption

These advantages are essential for running models in resource-constrained environments or high-volume enterprise systems, such as IBM Z.

- Model serving

  In the Model Serving phase, the trained model is compiled by using the IBM zDLC. The NVIDIA Triton Inference Server, optimized for IBM Z, serves the models. This configuration enables high-performance inference with PyTorch as the backend. The compute platform is Telum or Telum II processors, which are optimized for AI inference workloads on IBM Z.

- App infusion

  In the final stage, app infusion integrates the models with business applications that handle transactional workloads. These applications interact with the inference engine in real time through REST APIs.

  The applications, typically running on z/OS, use AI models to generate insights, predictions, and inferences that support business decision-making and operational efficiency.

For some use case examples that use AI Toolkit for IBM Z and LinuxONE, see 4.2, “Use case: An advanced multi-model AI framework for anti-money laundering” on page 40 and 4.3, “Use case: Credit risk assessment” on page 44.

2.2 IBM Cloud Pak for Data

IBM Cloud Pak for Data is a cloud-native data and AI platform that modernizes data management, analytics, and AI to help accelerate business outcomes. It takes advantage of IBM Z and LinuxONE features, including pervasive encryption, high availability, and virtualization through KVM or z/VM to help ensure robust support for data, analytics, and AI workloads.

This combination enables organizations to modernize data strategies with enhanced security and operational efficiency that is used particularly for mission-critical applications in industries such as finance and healthcare.

IBM Cloud Pak for Data offers significant advantages, including using the advanced Telum and Telum II embedded AI accelerator chip and the mainframe-grade quality of service. These capabilities help meet enterprise demands for high performance, scale, reliability, and unmatched security.

Note: For more information about IBM Cloud Pak for Data, see *IBM Cloud Pak for Data on IBM Z*, REDP-5695 and IBM Cloud Pak for Data.

2.2.1 Overview of services that are available

When you run Cloud Pak for Data running on Linux, you can perform the following tasks:

- Collect, store, ingest, and access data.
- Prepare and govern data.
- Run analytics, data science, and ML techniques on data.
- Build, train, deploy, and manage models.
- Automate and monitor the model lifecycle.
- Use open-source capabilities such as TensorFlow and Snap ML.

For more information about the services that are available with the latest IBM Cloud Pak for Data, see Planning for IBM Software Hub on IBM Z and LinuxONE.

Figure 2-8 shows all the components and services on IBM Cloud Pak for Data.

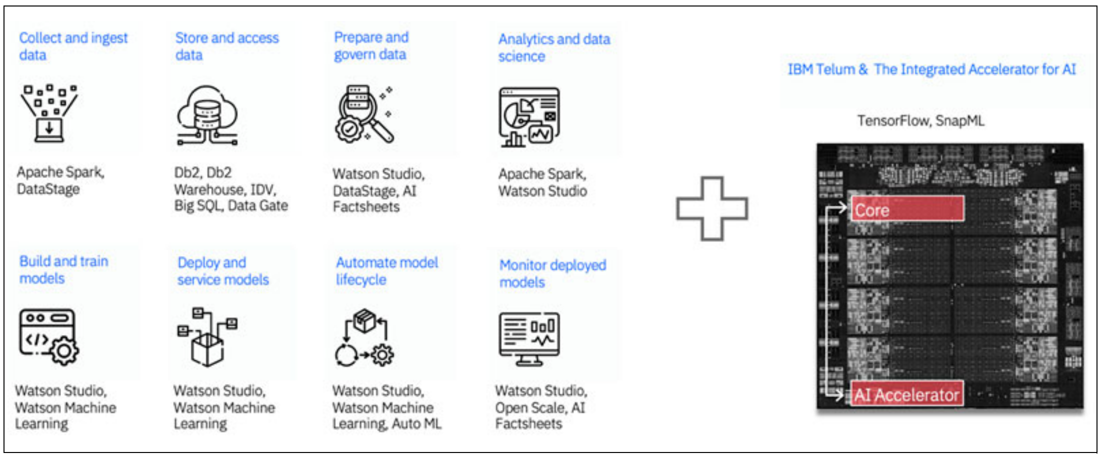

Figure 2-8 Available components and services on IBM Cloud Pak for Data

2.2.2 Getting started with Db2 Data Gate

IBM Data Gate with IBM Cloud Pak for Data gives you a modern hybrid cloud approach to access data originating on IBM Z. It offers an end-to-end solution to keep your data synchronized for hybrid cloud, analytics, and AI initiatives, while reducing cost and effort.

Db2 Data Gate keeps your Db2 for z/OS data available and synchronized from IBM Z sources to targets optimized on IBM Cloud Pak for Data. A log data provider captures changes from Db2 for z/OS and sends encrypted, consolidated updates to a log data processor running on IBM Cloud Pak for Data.

Db2 Data Gate can send updates to one or more targets, helping you reduce the complexity and cost of application development. The log capture process runs entirely on IBM Z Integrated Information Processor (zIIP) enabled, minimizing the impact on general processing on IBM Z. You can choose targets such as Db2 Advanced or Db2 Warehouse, based on your application requirements. These targets are fully optimized to deliver low latency and high throughput.

Figure 2-9 on page 21 shows the end-to-end flow of how Db2 z/OS data can be integrated and synchronized to IBM Cloud Pak for Data on the IBM Z architecture.

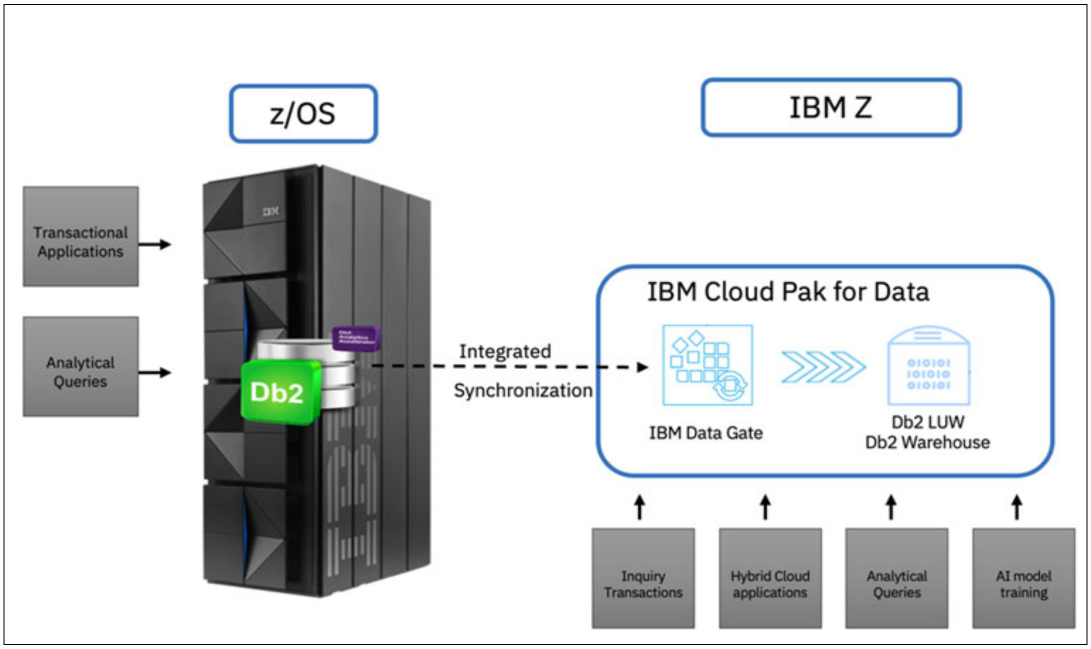

Figure 2-9 End-to-end flow

For more information, see Chapter 4, “Bringing it all together with the use cases” on page 37.

2.2.3 AI governance

AI governance refers to the frameworks, policies, and processes that help ensure the ethical, transparent, and responsible development, deployment, and use of AI systems. It includes guidelines for managing AI risks, helping ensure fairness, accountability, and regulatory compliance, while promoting trust and aligning AI outcomes with organizational and societal values.

In this section, you learn about three AI governance tools that can help you protect your data.

AI Factsheets

AI Factsheets is a core part of the AI governance solution that is available in Cloud Pak for Data and IBM watsonx.governance®. It captures data throughout a model's lifecycle, giving you and other stakeholders the information that is needed to evaluate models effectively.

Also, AI Factsheets increases transparency for auditors and regulators. The AI Factsheets catalog the data that is used to train the model, the frameworks and runtimes that are used in development, and the metrics that are collected during evaluation. Models are categorized by use case and organized by lifecycle stages, including development, testing, validation, and operation. If a model performs under its configured thresholds, you see an alert that shows the status of all models that are associated with the use case. You can also export AI Factsheets to various formats, such as PDF files, to support a wide range of stakeholders.

With AI Factsheets on IBM Z, you can ensure model governance and compliance by tracking and documenting the full history of AI models. You also need to monitor models in production to confirm that the models remain valid, accurate, and are free from bias or drift. You can achieve this goal by managing risk and promoting responsible AI through continuous monitoring of deployed AI models by using Watson OpenScale.

Watson OpenScale

IBM Watson OpenScale is a key component of the AI governance solution, which is also available in Cloud Pak for Data and IBM watsonx.governance. Watson OpenScale helps you monitor, evaluate, and analyze AI models with trust and transparency, so you can understand how AI models make decisions.

Watson OpenScale provides an enterprise-grade environment for AI applications, providing visibility into how your AI is built and used. Watson OpenScale enables you to operate and automate AI at scale with transparent and explainable outcomes.

You can monitor models in production to help ensure that models are valid, accurate, and aligned with intended goals without introducing bias or drift. This monitoring improves the quality and accuracy of AI model predictions. and helps you build high-quality models.

Watson OpenScale, together with AI Factsheets, helps you manage risk and support responsible AI.

IBM watsonx.governance

The IBM watsonx.governance toolkit is a governance toolkit that helps you direct, manage, and monitor AI activities across your organization.

The IBM watsonx.governance toolkit focuses on three key areas:

1. Lifecycle governance: Monitor, catalog, and govern AI models from any source throughout the AI lifecycle.
2. AI risk and security management: Identify and manage risks proactively by automating workflows and assigning accountability for controls that are associated with those risks.
3. Regulatory compliance: Help you meet regulatory requirements by converting regulations into submandates and enforceable policies.

With IBM watsonx.governance, you can manage risk and compliance across the entire AI lifecycle.

3

Building enterprise-ready artificial intelligence applications at scale

Enterprises have business-oriented requirements for their applications to support service-level agreements (SLAs). Ensure that artificial intelligence (AI) applications meet these requirements. Reliability, security, and scalability are essential characteristics for enterprise-ready AI applications and the infrastructure that support them.

AI at scale refers to deploying and operating AI systems across large, complex, or widespread environments to deliver significant impact, efficiency, or value. It involves moving beyond small-scale experiments or Proofs of Concept (PoCs) to implementing AI solutions that handle high volumes of data, users, or processes in real-world settings, often across an organization or industry.

This chapter guides you through the end-to-end process of building production-ready AI solutions at scale. It also describes the value of Linux in the context of AI applications.and concludes with key market use cases that run on Linux.

3.1 Reasons to deploy AI solutions on Linux on IBM Z and LinuxONE

As described in Chapter 1, “Introducing the convergence of artificial intelligence and enterprise Linux systems” on page 1, Linux enables enterprises to run Linux-based, mission-critical workloads in a reliable, secure, and scalable environment.

LinuxONE provides infrastructure for enterprise Linux workloads by using the core strengths of IBM Z. helping you meet enterprise demands.[1]

Modern AI applications fully use those core values (reliability, availability, serviceability, performance, and scale). When you deploy AI applications on the LinuxONE platform, they meet your enterprise’s strict SLAs and business regulations.

Another key value is an attractive total cost of ownership (TCO), which makes IBM Z and LinuxONE an economical choice for running AI workloads. Because AI workloads require significant hardware resources, you must consider TCO when selecting your infrastructure.

For AI-intensive workloads, IBM Z and LinuxONE provide an innovative on-chip AI processor that streamlines compute-intensive AI operations. Each IBM Z and LinuxONE computing chip includes a co-processor that transparently accelerates deep learning (DL) and predictive AI algorithms without requiring changes to your applications.

This AI acceleration is energy-efficient and effective across industries and use cases. Whether you are scoring business transactions for fraud, detecting anti-money laundering (AML) activity, or processing medical image recognition tasks, it delivers high performance without compromising efficiency.

The latest generation of IBM Z and LinuxONE builds on Telum technology and expands AI capabilities with Telum II and optional Spyre cards. These enhancements support multi-model AI composite solutions that combine traditional AI and generative AI models, helping you address a broader range of enterprise use cases.

Data is at the heart of every AI solution because it drives model development and training. LinuxONE inherits the IBM Z high-performance, data-driven architecture to deliver efficient and more streamlined data serving. You can deploy data serving solutions such as data lakes relational databases and SQL or No SQL databases on LinuxONE to bring AI processing closer to the data. This proximity reduces latency and improves model inference time.

You can embed AI solutions on the LinuxONE platform into transactional business applications and systems of record, and integrate them with data-serving and middleware products.

For a description of the data-serving solutions available on the platform, see Chapter 2, “Optimizing model serving solutions” on page 9.

Although IBM Z and LinuxONE provide a secure, high-performing, and scalable infrastructure for AI solutions, they also support a modern AI software ecosystem.This ecosystem enables you to access rich tools for AI application development, testing, deployment, and governance.

For high-performance AI model inferencing, you can choose between IBM enterprise software or open-source frameworks optimized for the IBM Z and LinuxONE architectures.

[1] Source: https://www.ibm.com/linuxone

When selecting infrastructure for deploying production-ready AI solutions, consider IBM Z or LinuxONE, which adhere to the following requirements:

- Availability: Serve data and AI solutions continuously.
- Reliability: Deliver data and AI solutions with 99.9999999% uptime.
- Scalability: Scale AI solutions horizontally and vertically to meet demand.
- Performance: Minimize latency and inference time by using built-in AIU co-processor and optimized software ecosystem.
- Security: Protect sensitive data, applications, and models to help ensure a secure production environment.
- TCO: Deliver all of the above in an economical and sustainable manner.

3.1.1 Reliable and available

Building a Red Hat OpenShift Environment on IBM Z, SG24-8515 highlights that Linux applications benefit from the highest level of availability in the industry. Starting with the z16 generation, IBM achieved 99.9999999% availability[2].

You achieve this level of availability through redundant hardware components, duplicated processor execution steps with integrity checking[3], and enterprise-grade firmware and middleware.

AI solutions, such as AI-powered fraud detection, deliver tremendous value to financial institutions and require a high level of reliability and availability. In addition to these technical capabilities, AI solutions on IBM Z and LinuxONE also emphasize content reliability, trustworthy AI, and AI governance. These features enable you to build scalable, business-critical, enterprise-ready AI applications.

3.1.2 Secure

As business-critical data and data that is protected under data protection laws are used in enterprise-ready AI applications, you must help ensure that this data remains secure always, which includes securing the data at rest (data-at-rest encryption), in transit over the network (data-in-transit encryption), and during active processing (data-in-use encryption).

IBM Secure Execution for Linux enables trusted execution environments that help you securely separate applications and data, minimize risks, and enhance security.

Support for quantum-safe cryptography prepares you for a post-quantum future, where quantum computers may break traditional encryption.[4]

With this broad set of capabilities, IBM LinuxONE and Linux on IBM Z enable you to implement confidential computing, helping ensure that your AI workloads remain secure, private, and compliant throughout their entire lifecycle.

[2] Source: https://newsroom.ibm.com/2024-02-06-New-IBM-LinuxONE-4-Express-to-Offer-Cost-Savings-and-Client-Value-through-a-Cyber-Resilient-Hybrid-Cloud-and-AI-Platform
[3] Source: *Leveraging LinuxONE to Maximize Your Data Serving Capabilities*, SG24-8518
[4] Source: https://www.ibm.com/linuxone/security

Note: IBM z17 can perform the following tasks:

- Process up to 35 billion encrypted requests per day with online transaction processing (OLTP) applications.
- Run OpenSSL bulk encryption with AES-256-XTS with up to 1.6x more throughput of a comparable x86 system.
- Scale I/O-intensive Linux applications and protect data at rest with up to 15 million read-only I/O operations per second and 7.5 million read/write I/O operations per second to an encrypted block device with FCP-attached storage.

3.1.3 Scalable

IBM LinuxONE and IBM Z use infrastructure components such as IBM Telum and IBM Telum II, enabling vertical and horizontal scaling. Because software licensing is a major cost factor, higher processor usage helps you reduce licensing costs when measured per core.

For more information, see Achieving Sustainability, Security, and Scalability with IBM LinuxONE 4.

Note: IBM z17 can perform the following tasks:

- Run an AI-infused OLTP workload on Red Hat OpenShift Container Platform, by using up to 5.6x fewer cores than a comparable x86 platform.
- Run Red Hat Enterprise Linux with KVM and deploy up to 3,000,000 NGINX containers.

3.1.4 AI acceleration

AI acceleration uses specialized hardware, software, or architectural optimizations to speed up AI workloads, including machine learning (ML) model training, inference, and data preprocessing. It enhances the performance, efficiency, and scalability of AI applications, especially in enterprise environments where reliability, security, and scalability are critical. AI accelerators are essential for processing the large volumes of data that is required by AI applications.

IBM Z and LinuxONE hardware architectures include dedicated, energy-efficient AI acceleration through on-chip components such as Telum and Telum II, and PCIe-based acceleration by using Spyre.

For more information about Telum II and Spyre, see New Telum II Processor and IBM Spyre Accelerator: Expanding AI on IBM Z and IBM LinuxONE.

3.1.5 Available AI frameworks and tools for Linux

As listed in Chapter 2, "Optimizing model serving solutions" on page 9, IBM provides a broad range of products for building and deploying enterprise-ready AI applications, supported by a wide variety of ISV software and open-source software.

> **Note:** For more information about distribution channels and how to implement the available resources in your own applications, see this GitHub repository.
>
> You can also find assets and guidance to help you use the Linux s390x architecture effectively.

Figure 3-1 shows a subset of AI frameworks available in the IBM Z and LinuxONE ecosystem that you can use to build and deploy enterprise AI solutions.

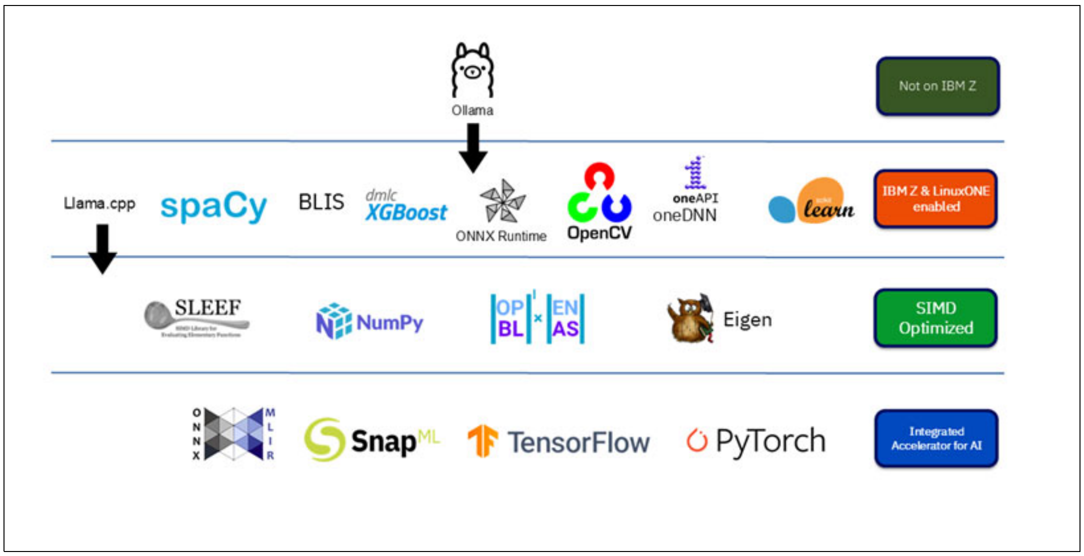

Figure 3-1 Subset of available AI frameworks for IBM Z and LinuxONE

The IBM z17 optimizes open-source AI frameworks by using the SIMD instruction set, the Integrated Accelerator for AI in IBM Telum and IBM Telum II, and the latest AI Accelerator, IBM Spyre, within the IBM Z and LinuxONE, s390x architecture[5].

To align with the high security standards of the IBM Z and LinuxONE infrastructure, IBM adds security checks to these AI frameworks and distributes them through its own registry.

For more information about obtaining open-source AI packages with optional IBM support, see Obtaining open source AI packages on IBM Z and LinuxONE.

5 Source: *IBM z17 (9175) Technical Guide*, SG24-8579

3.2 The AI lifecycle methodology

The AI lifecycle methodology includes seven key steps that help ensure that goals are met. Figure 3-2 shows an overview of the AI lifecycle methodology.

01	02	03	04	05	06	07
Define *your project goals*	Choose *a tool*	Prepare *the data*	Train *your model*	Deploy *your model*	Serve *model for inference*	AI Governance *monitor your model*
What do you want to find out? Do you have the data to analyze?	Pick the tools and entry point that matches your data and desired outcome	Refine the data Add the data as a project asset or in a data repository	Train the model with the data you supply, or fine tune an existing LLMs with your own data Let a model building tool choose estimators and optimize or choose your own	Move your model from development environment to a production environment	Make a deployed model accessible and usable for real time predictions or inference Exposing a model such as through a REST API	Monitor your model to evaluate the performance of models in production environments for bias and drift Provide explainability of features contributing to a prediction

Figure 3-2 AI lifecycle methodology

For details on developing and deploying an ML model on IBM z/OS, Linux on IBM Z, or LinuxONE using an AI lifecycle methodology, see Chapter 2., “Methodology and tools”, in *Optimized Inferencing and Integration with AI on IBM zSystems: Introduction, Methodology, and Use Cases*, REDP-5661.

Getting started with AI takes planning and time. IBM offers a no-charge discovery workshop to help you enable AI on IBM Z and LinuxONE and guide you through project planning and implementation.

You can use the following basic steps to enable AI in your enterprise through an AI Discovery Workshop.

1. AI on IBM Z and LinuxONE Discovery Workshop
2. Proofs of Concept (PoC) that use solution templates
3. Production planning

The following sections focus on the business perspective and best practices for successfully deploying an end-to-end enterprise AI solution. They begin with key personas and the design of a project plan. You can adapt this project plan to deliver a production-ready AI solution on LinuxONE or Linux on IBM Z. The chapter concludes with a secure end-to-end client implementation that you can use as a reference.

3.2.1 Starting the Proof of Concept: Personas

Typically, you should conduct a PoC within three months to validate your solution’s feasibility before committing to full-scale development or investment. A PoC serves as a low-risk, focused experiment to test critical assumptions and reduce uncertainty.

For an enterprise AI solution, clearly defining the business problem is essential. Delivering business value in an optimal, secure, reliable, and high-performance manner is the key performance indicator (KPI) of any AI project. All stakeholders should document and understand the business goals.

Gather key personas, including stakeholders and project team leads to clearly define the objectives and scope of the project:

1. Clearly define your objectives by identifying what you want to achieve and what business purpose the solution serves. Examples of some objectives include the following items:
   - Build a recommendation system.
   - Automate repetitive tasks.
   - Analyze data for faster decision-making (for example, fraud detection, insurance claims analysis, and AML).
   - Personalizing customer experience, such as using a chatbot.
   - Improve forecasting speed.
   - Optimize processes to reduce costs.
   - Helping ensure confidentiality through secure execution.
2. Clearly define the purpose of the PoC (for example, test a specific feature, technology, or process).
3. Identify key success criteria and measurable outcomes.
4. Create a document of understanding (DOU) that outlines the objectives, scope of the PoC, and the roles and responsibilities of all involved personas.

After you define a purpose and success criteria of the PoC with key stakeholders, you should involve additional team members to ensure a successful AI deployment.

Infrastructure teams, including IT, DevOps, and data engineers, often operate enterprise AI solutions. These personas might lack knowledge of business challenges, but typically understand infrastructure requirements and the unique components involved.

Application developers, data architects, or data scientist are business personas who work daily on solving business challenges, but often lack infrastructure knowledge.

Combine infrastructure personas, such as IT, DevOps, and data engineers, with business personas, including domain experts, product managers, and stakeholders, to help ensure project success, align with organizational goals, and deploy enterprise-ready AI effectively on IBM Z and LinuxONE. This collaboration bridges technical and business perspectives and enhances reliability, scalability, security, and TCO.

3.2.2 Project planning for a PoC and production implementation

Understanding business challenges and bringing together infrastructure and business personas are essential for any project, including a successful AI deployment on IBM Z or LinuxONE.

Document the key elements of the project plan to define the steps required to implement the PoC and support its migration to production.

Table 3-1 provides a sample of planning tasks that you should include in the DOU.

Table 3-1 Sample task list for PoC

Task	**Planned duration or date**	**Status**	**Responsibility**
1. Meet with key personas.	Pre-PoC	Done	Stakeholder Project lead
2. Project setup and staffing.	Pre-PoC	Done	All
3. Fulfill hardware and software requirements (refer to the installation roadmap).	Pre-PoC	Done	Infrastructure team
4. Provide a test license for the product.	PoC	Done	Product company
5. Download and installation of the product.	PoC	Done	Infrastructure team
6. Document and evaluate the results.	PoC	Open	All
7. Final PoC outcome presentation.	PoC	Open	All

After you complete the PoC and confirm that the outcome is acceptable, you can begin production implementation as a new project.

Use the PoC to evaluate how well the solution integrates with your existing infrastructure, identify any vulnerabilities or gaps, and adjust your production task list.

3.2.3 Example implementation of a PoC

This chapter describes an engagement with Client Y that you can use as a blueprint for implementing an enterprise-ready AI application at scale.

Client Y's application department aimed to implement AI-enhanced fraud detection. Previous engagements showed that integrating AI models with their traditional rule-based approach might improve results. To address business challenges such as rising fraud, the department decided to attend an IBM conference. They were aware that financial transactions were processed through IBM Z but needed more information.

After recognizing the benefits of implementing AI on IBM Z and LinuxONE, you understand that this infrastructure uniquely provides the reliability, security, and scalability that are required for enterprise-scale AI applications. As a result, the application department decided to proceed. They include both IBM and the infrastructure department in their planning. An AI Discovery Workshop is not required because the business challenge is clear and no further education on AI or the IBM Z and LinuxONE architecture is necessary.

To validate the feasibility of a production implementation, the team conducted two PoCs, as shown in Table 3-2 on page 31:

- The business PoC focused on developing an AI model for fraud detection that is enhanced by AI capabilities.
- The technical PoC focused on scaling enterprise-ready AI applications by evaluating whether a specific software solution on IBM Z can integrate with the existing infrastructure.

Table 3-2 Description of two PoCs

Business PoC	**Technical PoC**
► Train and evaluate AI. ► Choose features. ► Load tests. Can it be developed?	► Install the product. ► Test scalability and integration with the existing infrastructure. ► Measure resource requirements. Can it be used?

This method allows you to focus on the specific questions that must be answered before making a decision. At Client Y, the infrastructure and application teams work together to help ensure ongoing collaboration and alignment during the implementation of an AI-powered fraud detection solution.

After demonstrating the effectiveness of the solution on IBM Z, Client Y chose to deploy the solution at scale in collaboration with IBM.

Client Y followed this process for their PoCs in the following way:

- The application department identified the business challenge.
- The application department engaged the infrastructure department and IBM.
- IBM was involved to address challenges and answer questions.
- A DOU and project plan were created.
- The team had regular meetings throughout the project.

3.3 Examples of use cases

This section introduces use case ideas that span both traditional AI (predictive AI) and generative AI, enabling cross-engagement through multi-model AI architectures.

For technical insights into implementing a real-time, in-transaction scoring use case, see Chapter 3., "Real-time in-transaction scoring use case scenarios", in *Optimized Inferencing and Integration with AI on IBM zSystems: Introduction, Methodology, and Use Cases*, REDP-5661. For descriptions of more predictive AI use cases beyond the ones that are covered here, see Chapter 4., "Other use case scenarios", in *Optimized Inferencing and Integration with AI on IBM zSystems: Introduction, Methodology, and Use Cases*, REDP-5661.

Figure 3-3 highlights several AI use cases on IBM Z and LinuxONE that benefit from the enhanced capabilities of Telum II and Spyre.

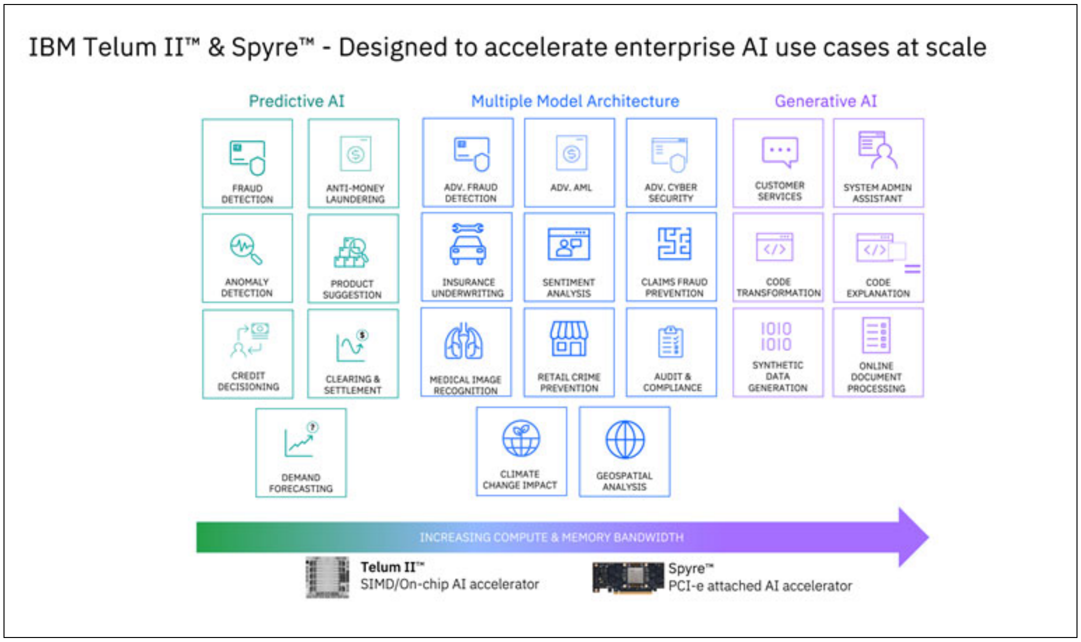

Figure 3-3 Current AI on IBM Z and LinuxONE use case landscape

Note: Figure 3-3 does not show all possible use cases. For more information about an AI on IBM Z and LinuxONE Discovery Workshop, contact your IBM representative or email the AI on IBM Z and LinuxONE team at aionz@us.ibm.com.

3.3.1 Predictive AI

Predictive AI systems analyze historical data and ML) and DL models to predict future events or trends. For more information, see Chapter 1, "Foundations of artificial intelligence", in *Optimized Inferencing and Integration with AI on IBM zSystems: Introduction, Methodology, and Use Cases*, REDP-5661.

Because predictive AI models are lightweight and effective for decision making, you can use them in real-time AI-infused transactions.

In time-critical transactions where milliseconds matter, external API calls often fail to meet SLAs. The Telum and Telum II processors each include an AI acceleration unit on each chip and are optimized for workloads that require rapid, embedded AI processing, which make them ideal for the following use cases:

- Low-latency requirements
- A secure environment due to sensitive data
- No external API call because the transaction already runs on IBM Z or LinuxONE

Predictive AI is suitable for use cases that require trustworthy AI, as mandated by legal frameworks such as the EU AI Act. Although you can achieve trustworthy AI with generative AI, it is simpler to implement by using predictive AI models.

Note: IBM Z products received several awards in the area of trustworthy AI[a]. The IBM watsonx.governance toolkit provides trustworthy AI capabilities for generative AI[b]. AI Factsheets and OpenScale, available in IBM Cloud Pak for Data, support trustworthy AI for predictive AI models, available on Linux. For more information, see Chapter 2, "Optimizing model serving solutions" on page 9.

a. In 2024, IBM Machine Learning for IBM z/OS (MLz) received the AI Excellence Award from the Business Intelligence Group for its innovative and trustworthy AI capabilities. For more information, see Machine Learning for IBM z/OS wins AI Excellence Award from Business Intelligence Group. In 2025,the solution also earned the prestigious iF Design Award for its user-centric design and seamless integration of AI into transactional workloads on IBM MLz. For more information, see IBM Machine Learning for IBM z/OS Enterprise Product: These awards highlight IBM's continued leadership in delivering secure, reliable, and enterprise-grade AI solutions.

b. Source: `https://www.ibm.com/products/watsonx-governance`

3.3.2 Generative AI

Generative AI refers to a type of AI that creates new content, such as text, images, music, or even code, based on the data that it has been trained on. You can use generative AI for advanced AI applications that require content generation or synthesis.

Here are some examples of using generative AI:

- System administrative assistant

  When your infrastructure team faces an increased workload but lacks sufficient personnel, an AI assistant can help you improve efficiency. IBM watsonx Assistant for IBM Z addresses this need by providing a dedicated solution that enables you to complete system administration tasks directly from a chat interface. This capability helps streamline operations, reduce manual effort, and improve response times.

  > **Note:** For more information about how IBM watsonx Assistant for IBM Z can help you, see watsonx Assistant for IBM Z.

- Code explanation and optional code transformation

  To help you understand and modernize code, IBM provides a specialized product for mainframe infrastructure: IBM watsonx Code Assistant for IBM Z. This product also includes an optional feature that helps you transform code into Java.

  > **Note:** For more information about Code Assistant™ for IBM Z, see watsonx Code Assistant for IBM Z.

- Synthetic data generation

  You need high-quality data to test and build AI models, which IBM Synthetic Data Sets addresses this need.

  > **Note:** For more information about IBM Synthetic Data Sets, see IBM Synthetic Data Sets.

3.3.3 AI multiple-model architecture

Because of their lightweight design, predictive AI models deliver high performance but struggle to process unstructured data or explain decisions in detail. Generative AI can complement predictive AI in these areas. You can use multi-model AI architectures when you need more detail if the transaction is not time critical, or the transaction is time critical, but you can review or explain it afterward.

The following are two industry examples of using multi-model AI architectures:

- Insurance industry

  Consider an enterprise-ready AI model for processing hail damage payment claims. A predictive AI model efficiently handles structured data, such as payment amounts or no-claim bonuses, to make payment decisions. However, it cannot process unstructured data, such as appraiser photos or police reports. A generative AI model can analyze this unstructured data, convert it into structured data, and feed it to the predictive AI model to enhance decision making.

- Financial Industry

  In a fraud detection scenario, predictive AI models are essential for low-latency decisions, such as approving or declining payments. Generative AI adds value after the transaction by generating insights or explanations.

  You can automatically send detailed explanations for declined transactions to clients.

3.3.4 AI Security

In the rapidly evolving landscape of AI, helping ensure robust security measures is essential. IBM helps you protect AI systems and data through its AI Security initiatives, which use innovative technologies and practices.

This section describes three key AI security initiatives:

- IBM Secure Execution provides enhanced protection for sensitive workloads.
- Confidential AI helps ensure privacy and security in AI processes.
- IBM Synthetic Data Sets offer secure and reliable data for training AI models.

Together, these initiatives help you advance AI security and maintaining trust in AI-driven solutions.

IBM Secure Execution

Enterprise-scale AI applications with strict data security requirements can benefit significantly from IBM Secure Execution for Linux.

IBM Secure Execution for Linux is a security technology that was introduced with IBM z15® and LinuxONE III. It protects the data of workloads running in a KVM guest from inspection or modification by the server environment. As a result, no hardware administrator, KVM code, or KVM administrator can access the data as a guest that starts in secure-execution mode.

In the healthcare industry, you can use IBM Secure Execution to protect sensitive data, including X-rays. These highly sensitive files require secure storage and execution, which you can manage within a secure enclave. IBM Z and LinuxONE provide the infrastructure you need to support this level of security.

For more information about IBM Secure Execution, see What is IBM Secure Execution?

Confidential and secure AI

As you integrate predictive and generative AI, you face challenges in deploying these innovations without invalid configurations, security vulnerabilities, or other risks that might lead to data protection breaches or compliance violations. This situation is where IBM's approach to confidential AI helps ensure the privacy and security of your AI processes and data.

By following IBM's framework for securing generative AI initiatives, you should consider approaches that protect your datasets, models, and use of trained and deployed models. Also, evaluate strategies that protect your AI infrastructure and establish strong AI governance.

IBM's confidential and secure AI strategy uses confidential computing technologies to protect data during processing. These technologies rely on hardware-based enclaves that keep sensitive information encrypted and isolated. By using secure computing technologies, you can protect sensitive data and AI models from threats and attacks while maintaining trust.

With secure AI, you can train models by using artificial data, monitor the models for fairness and bias, and generate governance reports to support transparency.

For more information about the IBM framework for securing generative AI, see Introducing the IBM Framework for Securing Generative AI.

IBM Synthetic Data Sets

IBM Synthetic Data Sets are a family of enterprise-grade, artificially generated, datasets that help you enhance predictive AI model training and large language models (LLMs) for IBM Z and IBM LinuxONE clients, ecosystems, and independent software vendors.

You can download these pre-built datasets as comma-separated values (CSVs) and data definition language (DDL) files. They are simple to use and compatible with databases, spreadsheets, hardware platforms, and standard AI tools. These datasets also highlight IBM's industry expertise in the financial services sector without using real client seed data, helping you avoid security concerns that are related to personally identifiable information (PII).

IBM Synthetic Data Sets help you train and enhance predictive models and composite AI methods. You can deploy these models to IBM Z and LinuxONE by using inferencing tools such as IBM MLz, AI Toolkit for IBM Z and IBM LinuxONE, and IBM Cloud Pak for Data on IBM Z.

The IBM Synthetic Data Sets family includes the following options:

- IBM Synthetic Data Sets for Payment Cards
- IBM Synthetic Data Sets for Core Banking and Money Laundering
- IBM Synthetic Data Sets for Homeowners Insurance

These datasets help you simulate real-world scenarios and protect sensitive information, making them ideal for training and testing AI models in regulated industries.

For more information about IBM Synthetic Data Sets, see *IBM Synthetic Data Sets*, REDP-5748.

4

Bringing it all together with the use cases

This chapter describes two solutions that are suitable for IBM Z and LinuxONE. You can apply the approach and solution architecture to any industry or sector.

This chapter begins by introducing several IBM Z and LinuxONE frameworks that accelerate applications' access to data because data is crucial for building and training artificial intelligence (AI) models. The first use case demonstrates how to embed these data solutions within your AI framework.

The first solution scenario uses a multi-model AI approach to detect fraud in home insurance claims. This scenario uses replicated Db2 for z/OS transactional data, which you access through IBM Data Gate. It augments the data with contextual embedding features that are derived from large language models (LLMs). You use the merged data as input for a neural network framework to detect potential fraud. You can apply this multi-model AI approach to other industries. This combination of predictive and generative AI helps you achieve higher precision and accuracy.

4.1 Use case: Advanced multi-model AI framework for home insurance fraud detection processing

This section describes an advanced multi-model AI framework that combines traditional and generative AI approaches to detect potential fraud in high-volume home insurance claims with high precision and accuracy.

This integrated approach enables you to use generative AI to process unstructured narrative data from claims and derive meaningful context embeddings that enhance structured insurance claims data.

IBM Z, equipped with an IBM Spyre AI Accelerator, provides an ideal AI infrastructure for implementing this combined model. You can apply this architectural approach, which combines generative AI with predictive AI, across many industries and domains, including the insurance sector. The solution includes IBM Data Gate technology, described in this chapter, which replicates structured Db2 for z/OS table data into the lakehouse.

The business impact of deploying the entire solution on IBM Z:

- Shortening the insurance claims lifecycle
- Increasing customer satisfaction by prioritizing claims based on AI recommendations
- Boosting the efficiency of the insurance claims team
- Delivering high confidence in processing and scoring 100%of transactions at scale.

4.1.1 Overview, challenges, and needs for fraud detection in insurance

According to a study from Celent, fraud caused an estimated $385 billion in global losses to the banking, cards, and payments sectors in 2021. The insurance sector also experienced significant fraud-related losses. PricewaterhouseCoopers (PwC) estimated these losses at $308.6 billion in the United States alone[1].

As financial crime becomes more complex, you must adopt advanced methods to detect fraud, adapt to modern challenges, and strengthen your detection capabilities. Although traditional predictive AI algorithms offer a certain level of accuracy, combining them with generative AI frameworks creates opportunities to improve fraud detection and prevention.

A more modern approach uses an advanced multi-model AI system that combines predictive AI and generative AI to enhance fraud detection in insurance claims. You can deploy this solution on-premises on IBM Z to address key industry challenges such as service-level agreements (SLAs) for insurance transaction processing, secure handling of sensitive data, and compliance with industry standards. IBM Z provides an optimal infrastructure for scaling resources dynamically. The Telum II processor and Spyre AI Accelerator on IBM Z support fast and secure AI inferencing.

[1] Source: https://www.pwc.com/gx/en/industries/financial-services/publications/financial-crime-in-the-insurance-industry.html

4.1.2 Advanced multi-model AI insurance claims fraud detection: A reference architecture on IBM Z and LinuxONE

This architecture contains the following main components:

- Db2 z/OS
- Lakehouse
- Data Gate
- IBM watsonx.ai®
- Triton Inference Server

Data is at the heart of this process, and insurance claims processing involves several steps. This section explores how you collect, curate, and merge data, to improve AI model development and training (see 4.2, “Use case: An advanced multi-model AI framework for anti-money laundering” on page 40).

You replicate a Db2 for z/OS table containing structured claims data to an IBM watsonx.data® lakehouse by using the high-speed Data Gate synchronization protocol. The system writes the data to the Apache Iceberg open table format for further processing.

The IBM watsonx.data lakehouse merges structured claims data from Db2 with contextual embeddings that are extracted by using an LLM. This combined data supports AI model development and training.

The Data Gate protocol automatically detects updates to the source table and replicates them to the lakehouse, helping ensure that you work with the latest data.

Next, you extract features such as claim descriptions and narratives from unstructured claims data by using an LLM running on IBM Z. You can accelerate this process with IBM Z Spyre cards.

The extracted features are converted into BERT encoder embeddings with inferred sentiments, helping you capture the nuances and urgency of each claim, especially those claims that are emphasized in the claim description.

You use the extracted features for downstream tasks such as classification or regression to rank claims effectively. Predictive AI model scoring identifies high-risk patterns and flags claims with characteristics of potential fraud.

You deploy the model by using Triton Inference Server, which enables efficient, scalable AI inferencing to quickly identify fraudulent home insurance claims, which helps your organization prioritize efforts to improve customer service and optimize resource utilization.

The multi-model AI approach combines LLMs for deep text understanding with neural networks for pattern recognition in structured data. This combination significantly improves fraud detection accuracy. IBM Z hardware acceleration enhances overall performance and scalability.

Figure 4-1 shows an overview of the advanced Multi-Model AI insurance claims fraud detection on IBM Z.

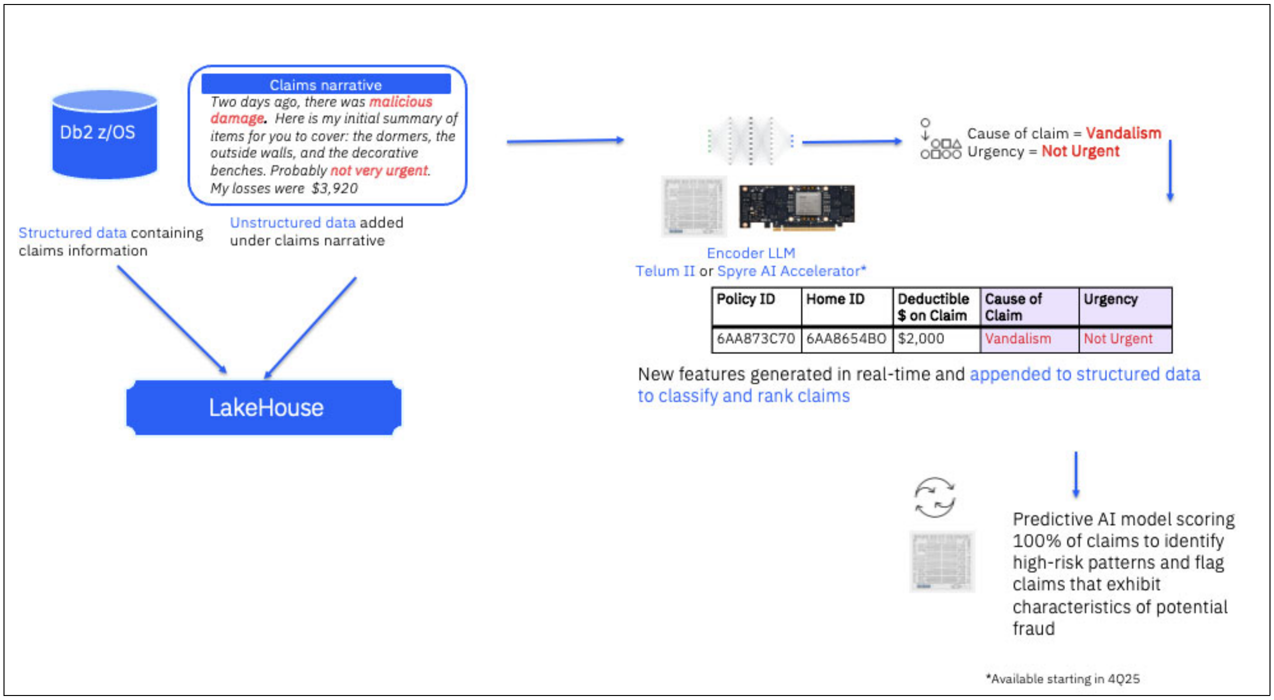

Figure 4-1 Advanced Multi-Model AI insurance claims fraud detection on IBM Z

4.2 Use case: An advanced multi-model AI framework for anti-money laundering

This section describes ran advanced real-time multi-model AI framework that combines traditional and generative AI approaches to prevent money laundering with high precision and accuracy. This approach enhances illicit activity screening.

This integrated multi-mode AI framework enables you to use generative AI to enrich traditional predictive AI with real-time Know-Your-Customer (KYC) verification. By using LLMs, it processes unstructured data, including social media, news, online forums, and other sources.

Real-time multi-model AI inferencing on IBM Z improves overall system accuracy, helping financial institutions and regulators gain deeper insights and detect fraud more effectively. IBM Z, equipped with a Spyre AI Accelerator, provides an ideal AI infrastructure for implementing this combined model.

You can apply this architectural approach, which integrates generative AI and predictive AI, across a wide range of industries, including the financial sector.

The business impact of deploying an entire solution on IBM Z:

- Detects illicit activity with high precision.
- Enables real-time processing to reduce regulatory compliance costs.
- Preserves economic transparency and integrity.

4.2.1 AML implementation: Overview, challenges, and requirements

As money laundering grows more sophisticated, effective prevention and response require a robust AI-driven approach. According to a Celent study, regulators frequently impose heavy fines on banks for inadequate anti-money laundering (AML) programs. High false positive rates in many AML operations place a severe burden on financial institutions.

If left unchecked, money laundering enables global crimes such as drug trafficking, human smuggling, and terrorism.

A modern approach uses an advanced multi-model AI system that combines predicative AI and generative AI for AML detection and verification. Deploying this solution on-premises on IBM Z addresses key industry challenges, including SLAs for financial transaction processing, secure execution of SPI and other sensitive data, and compliance with industry standards.

IBM Z provides the optimal infrastructure for scaling resources dynamically in a sustainable way. Telum II and Spyre Accelerator on IBM Z enable fast and secure AI inferencing.

4.2.2 Reference architecture of the Multi-Model AI AML framework on LinuxONE

AML screening is complex and involves multiple steps. An internal IBM benchmark shows that combining predictive and generative AI models improves precision and accuracy in detecting illicit activities. Standard AML processing includes AI-based transaction and route monitoring. LLM-infused AML enhances this monitoring by using unstructured data sources such as existing KYC data, social media activity and inactivity, online forums, and more.

This innovative approach enhances accuracy and provides new insights based on all available data. The AI on IBM Z toolkit combines containerized open-source inferencing platforms to provide an inferencing layer that supports AML implementation on LinuxONE.

You can review an advanced multi-model AI framework that screens financial transactions to prevent money laundering:

1. In Figure 4-2, the AI model identifies transaction patterns based on historical payment data that uses graph algorithms. The AI model compares hidden relationships and interactions in real time against both common and uncommon AML patterns.

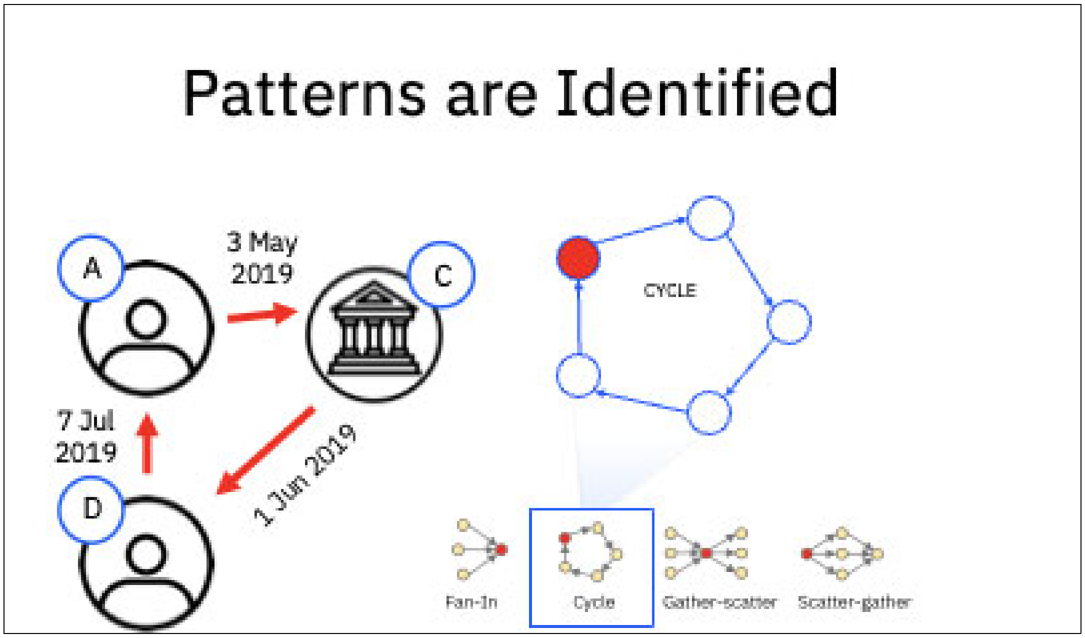

Figure 4-2 A process of identifying patterns and relationships of financial transactions

2. The AI model evaluates all key features on a weighted basis, including suspicious transaction amounts, the number of transactions in each cycle, and other anomalies compared with prior activity (Figure 4-3 on page 43).

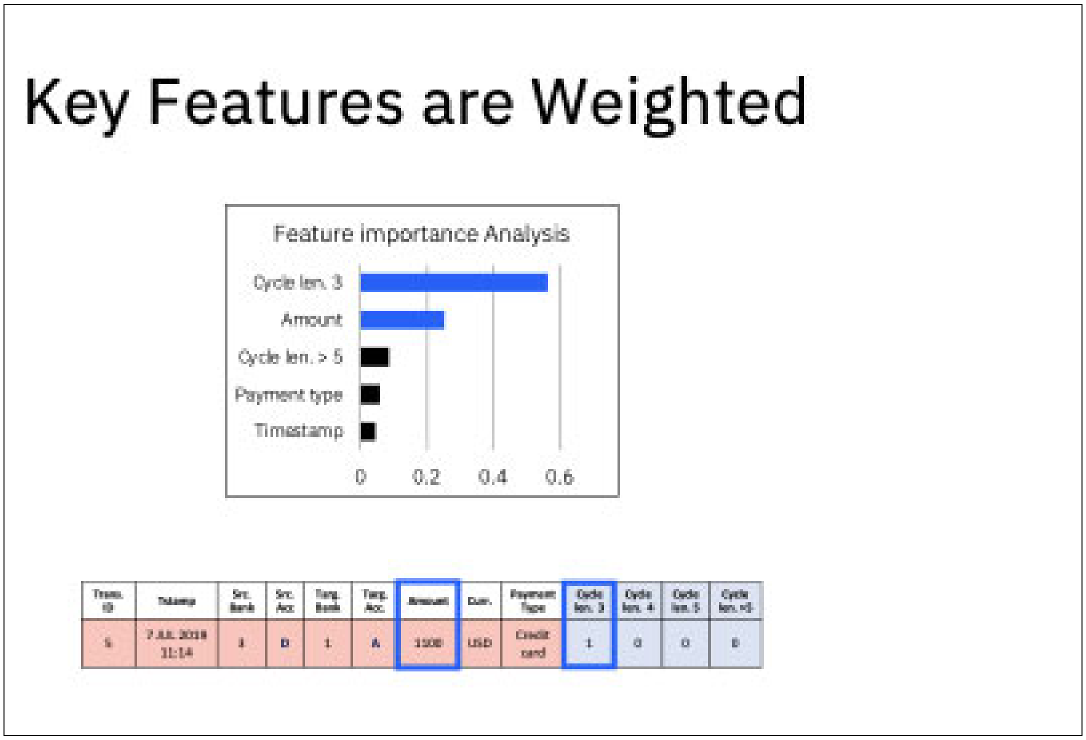

Figure 4-3 Key features are weighted

3. In Figure 4-4, you enhance the AML screening process by infusing it with generative AI sentiment analysis that uses LLMs. This step processes large volumes of unstructured data, such as social media and KYC information, to uncover hidden patterns that might indicate illicit transactions.

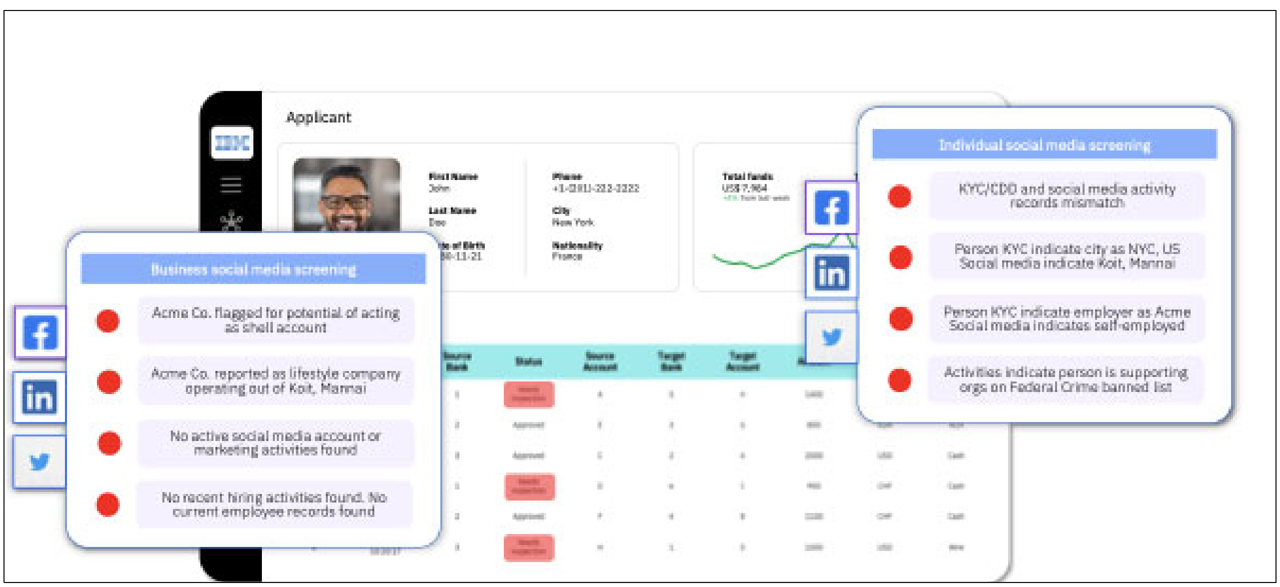

Figure 4-4 Unstructured data processing

4. The AI model incorporates a wide range of variables and insights from previous steps and becomes more refined with each new transaction. It then generates an accurate, explainable recommendation to determine the likelihood of illicit activity.

Infusing the framework with sentiment analysis that uses an LLM significantly enhances accuracy. Fine-tuning the LLM further improves accuracy.

The system uses LLMs to make logic-driven predictions on unstructured data, enhance illicit activity screening, and improve model accuracy.

4.3 Use case: Credit risk assessment

This use case shows how you can use AI to enhance credit risk assessment for financial institutions. It integrates data management, machine learning (ML), and real-time decision-making to improve operational efficiency, prevent fraud, and enhance the customer experience.

Data management is essential. This example uses IBM Db2 for z/OS to integrate and manage structured and unstructured data. The AI-driven assessment uses IBM ML and Snap ML to train and deploy models. Real-time decision-making enables instant credit approval or rejection based on AI inference.

Analytics dashboards help you monitor trends and retrain models to continuously improve accuracy.

End-to-end flow

This section presents an example of an AI architecture for a credit risk assessment system on IBM LinuxONE. The system relies on high-performance data storage to manage large volumes of data efficiently.

Figure 4-5 shows the entire end-to-end flow.

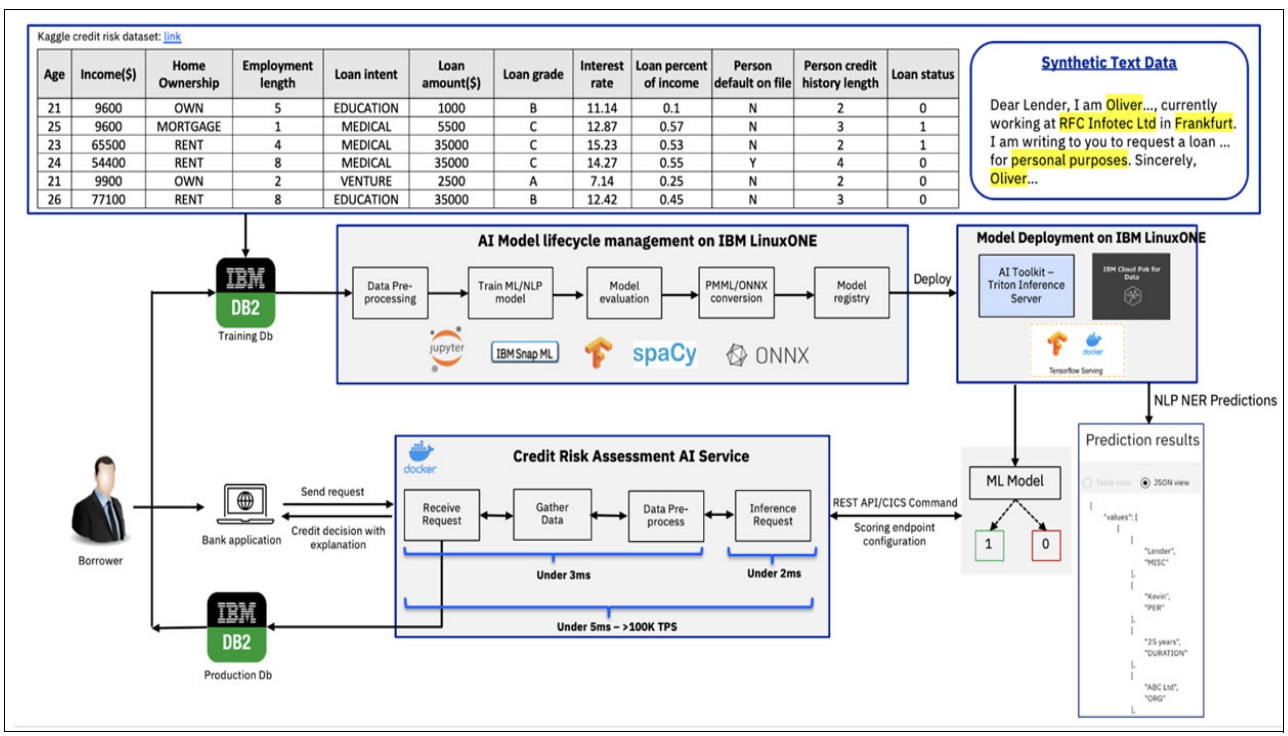

Figure 4-5 AI architecture of credit risk assessment system on IBM LinuxONE

Figure 4-6 shows a sample of synthetic text data that you can use to train the AI system. Synthetic data augments the existing dataset, improving the model's ability to generalize and handle diverse inputs. By using synthetic data, you enable the system to rely on historical credit data instead of personal data.

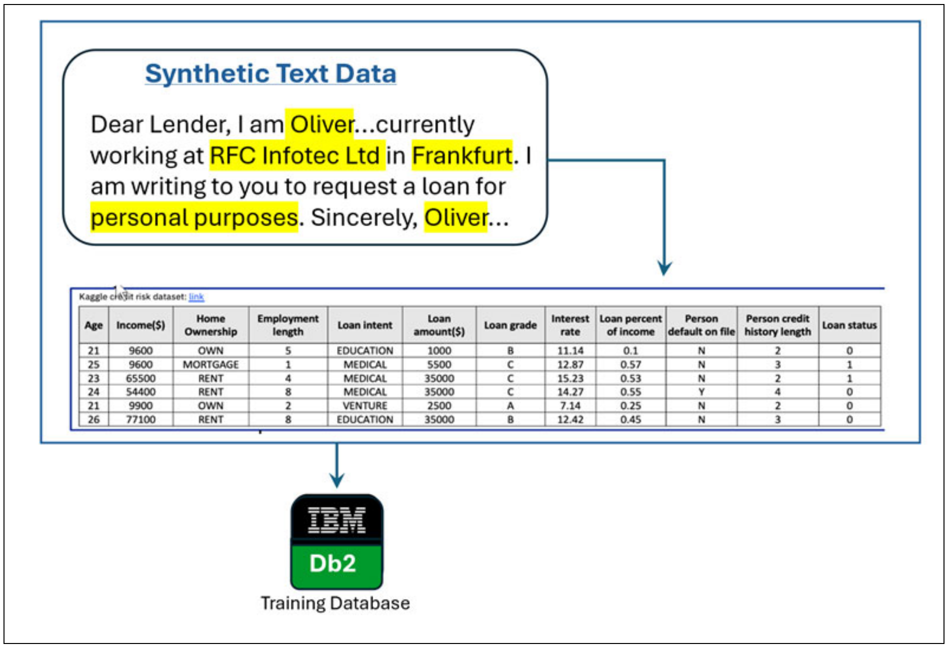

Figure 4-6 Synthetic text data

You can use IBM Synthetic Data Sets as your source of synthetic text data. These datasets are artificially generated to mimic real-world data while preserving privacy and security. They are designed for testing, ML model development, and research, offering realistic patterns and structures without exposing sensitive information.

In this example, the IBM Db2 training database stores the raw dataset that is used for training.

Figure 4-7 shows how you clean and prepare the dataset by using a Jupyter Notebook. You apply ML and natural language processing (NLP) models by using IBM Snap ML, a library for scalable ML, and spaCy, an NLP library. Snap ML handles structured data, such as numerical features like income and loan amount, while spaCy processes unstructured text such as synthetic text data.

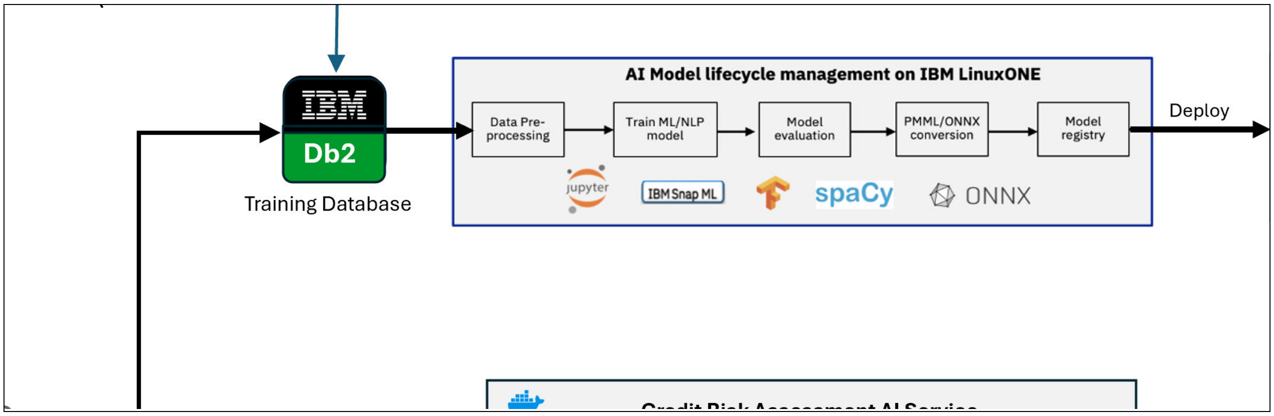

Figure 4-7 AI model lifecycle management on IBM LinuxONE

During model evaluation, you assess the trained model for accuracy and precision. You then convert the model to Predictive Model Markup Language (PMML) or Open Neural Network Exchange (ONNX) format for deployment.

Figure 4-8 shows how the converted model is stored in a registry for versions and deployment.

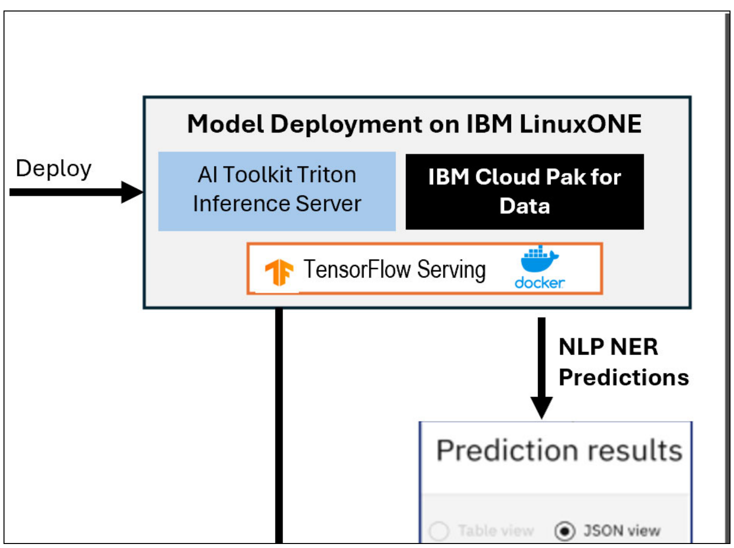

Figure 4-8 Deployment phase of the AI model lifecycle

During deployment, you make the trained model available for real-time predictions. You use the AI Toolkit for IBM Z and LinuxONE Triton Inference Server to serve the trained ML model for credit risk prediction and help ensure fast, scalable inference on IBM LinuxONE.

You use IBM Cloud Pak for Data to manage the AI lifecycle, including data preparation, model training, and deployment. It also manages model versions, monitoring, and governance to help ensure that the deployed model remains reliable and compliant with enterprise standards.

You use TensorFlow Serving to deploy TensorFlow models in production. It works with Triton Inference Server to support multiple model types.

You use Docker containers to package the model and its dependencies, helping ensure consistency and portability across environments on IBM LinuxONE. This approach supports seamless deployment and scalable inference services.

When a borrower submits a request through the bank application, data collection and integration begin. At this stage, borrower details, historical data, and alternative sources are unified. You use IBM Db2 for z/OS profiling to ensure data quality and consistency. This profiling helps detect issues such as a negative income or missing loan status, which can distort the AI model's predictions.

Figure 4-9 shows the prediction results of the credit risk assessment system. This system provides two types of predictions:

- ML Model Predictions: The ML model outputs a binary classification (1 or 0) indicating credit risk where 1 represents high risk of default and 0 represents low risk.
- NLP NER Predictions: The NLP model, which is trained with spaCy as shown in the larger diagram, processes text data and outputs named entities.

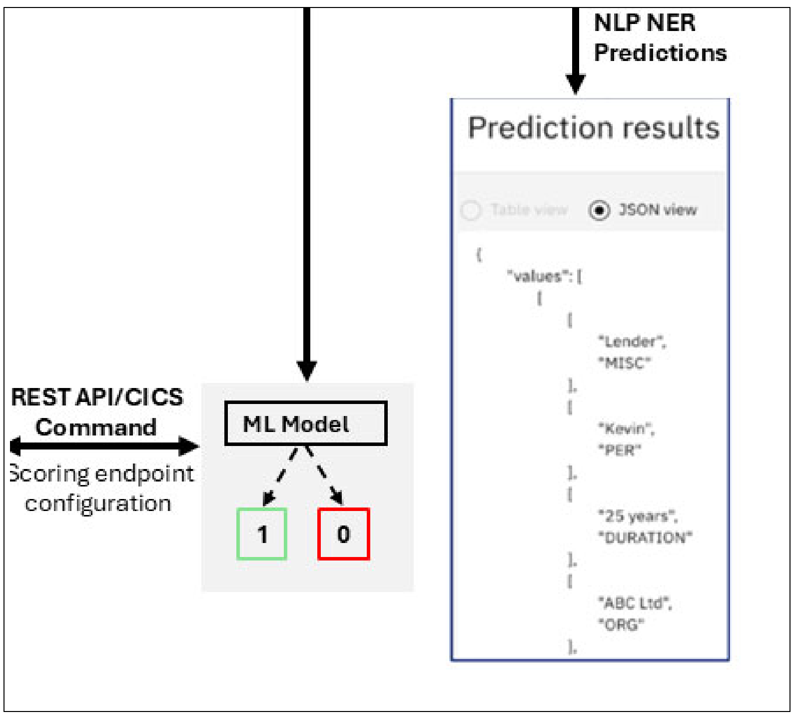

Figure 4-9 Prediction results of the credit risk assessment system

You train ML models, such as Random Forest and Boosting Classifiers, on this data to predict loan status. These models achieve high accuracy based on metrics such as ROC AUC scores. You deploy the trained models on IBM Machine Learning for z/OS to create scoring endpoints for real-time inference.

Figure 4-10 shows how you process a borrower's loan application in real-time, deliver a credit risk assessment, and return a decision to the banking application.

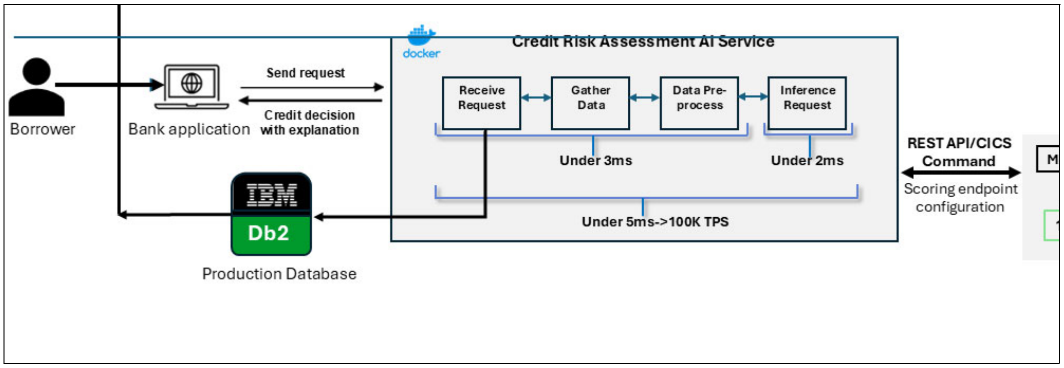

Figure 4-10 Delivering the credit risk assessment

A borrower submits a loan application through the banking application, and the system predicts whether to approve or reject the request. Explanatory tools help you enhance transparency during this process. The banking application sends a request to the credit risk assessment AI service.

The credit risk assessment AI service receives the borrower's data, retrieves relevant information from the production database, and prepares it for inference. Figure 4-9 on page 47 shows how the ML model processes the data and predicts the credit risk. The system returns a decision (approve or deny) with an explanation to the banking application.

You maintain model accuracy through continuous learning by retraining on production data to handle data drift. This end-to-end process improves operational efficiency, reduces fraud, enhances the customer experience, and helps you make data-driven decisions.

4.4 Overview of additional use cases

Many other industries and use cases are also suitable for Linux on IBM Z and LinuxONE technologies. The following industry use cases illustrate business scenarios, business impact, and example solutions that run on Linux, including Linux on IBM Z and LinuxONE.

Banking

This section covers the following topics:

- Know-Your-Customer-based short loans
- Improved operations for credit card applications

Know-Your-Customer-based short loans

Business scenario: A client wants to predict the likelihood that a borrower repays point-of-sale loans. This information enables users to use their banking relationships as a form of credit account overdraft.

Business impact: The client can now generate near real-time AI predictions of a borrower's likelihood to repay, enabling an extended revenue stream through short-term loan usage fees.

Example solution:

- Data or application: Db2 for Linux, UNIX, and Windows (Db2 LUW)
- AI platform or framework: IBM Cloud Pak for Data

Improved operations for credit card applications

Business scenario: A client wants to move fraud detection models to the IBM Z environment to improve scalability, availability, and response time issues that are limited by the existing platform.

Business impact: The client can avoid the cost of AI outages and help ensure that AI functions have the scalability and resiliency that are required for enterprise operations alongside other critical infrastructure.

Example solution:

- Data or application: MongoDB
- AI platform or framework: TensorFlow

Insurance

This section covers the following topics:

- Claims image analysis
- Operational efficiency and costs

Claims image analysis

Business scenario: A client wants to predict whether an image of minor car accident damage represents a valid claim and whether it matches previously paid claims.

Business impact: The client can now improve assessment speed and reduce the time that is spent on manual follow-ups by prioritizing cases based on risk assessments.

Example solution:

- Data or application: PostgreSQL
- AI platform or framework: PyTorch, ONNX, or deep learning compiler (DLC) through the AI Toolkit for IBM Z and LinuxONE

Operational efficiency and costs

Business scenario: A global health insurer wants to analyze large volumes of medical records to predict disease risk while maintaining compliance with HIPAA regulations.

Business impact: The client can now quickly identify high-risk scenarios and reduce data center hardware footprints and energy consumption, resulting in reduced costs for space and power.

Example solution:

- Data or application: MongoDB
- AI platform or framework: PyTorch, ONNX, or DLC through the AI Toolkit for IBM Z and LinuxONE

Government

This section covers the following topics:

- Private data chatbot
- Reducing the risk of errors and operational costs

Private data chatbot

Business scenario: A government client wants to offer automated services for federal benefits management. The goal is to enable chatbot-driven automation of repetitive tasks and provide personalized user experiences to help resolve questions and complete basic account updates involving private data.

Business impact: The client can now automate manual, repetitive tasks to reduce operational costs and improve user satisfaction, while still meeting regulatory standards.

Example solution:

- Data or application: MongoDB
- AI platform or framework: TensorFlow

Reducing the risk of errors and operational costs

Business scenario: A federal client wants to incorporate family relationships into immigration and customs processes as part of predictive risk analysis for criminal concerns. Even a moderately accurate model that uses ML and NLP can reduce manual checks across names, misspellings, and languages.

Business impact: The client can reduce the cost of manual reviews and reduce space and power requirements.

Example solution:

- Data or application: MongoDB
- AI platform or framework: TensorFlow

Wholesale distribution

This section covers the following topics:

- Variable pricing trends
- Improving partner onboarding

Variable pricing trends

Business scenario: A client wants to predict regional pricing optimization opportunities by using incoming supply volumes, seasonal trends, and types of goods.

Business impact: The client implements pricing contracts that include a variable range, or "market value," for key product types. Market value is determined through an automated governance model that remains auditable for regulatory compliance.

Example solution:

- Data or application: MongoDB
- AI platform or framework: PyTorch, ONNX, or DLC through the AI Toolkit for IBM Z and LinuxONE

Improving partner onboarding

Business scenario: A client wants to predict whether a new vendor or product is a good fit for the partner network by identifying similar vendors or complementary products based on KYC information for buyers.

Business impact: The client runs a batch workload that analyzes past transactional data and highlights similar or dissimilar vendor patterns across buyers and regions.

Example solution:

- Data or application: PostgreSQL
- AI platform or framework: PyTorch, ONNX, or DLC through the AI Toolkit for IBM Z and LinuxONE

Education

This section covers the following topics:

- AI training classes
- Reducing the risk of loss

AI training classes

Business scenario: A client wants to offer training classes to improve AI prediction models for fraud detection, AML, and other high-transaction or sensitive data use cases in an optimized and sustainable way.

Business impact: The client takes advantage of scalability by using industry-standard Linux applications to run sample fraud detection and AML models with synthetic datasets.

Example solution:

- Data or application: PostgreSQL
- AI platform or framework: PyTorch, ONNX, or DLC through the AI Toolkit for IBM Z and LinuxONE

Reducing the risk of loss

Business scenario: A client aims to improve student satisfaction with payroll and student information systems by using AI models to detect incorrect information caused by unexpected values.

Business impact: The client significantly reduces errors and manual follow-ups, and improves student satisfaction.

Example solution:

- Data or application: Db2 LUW
- AI platform or framework: IBM Cloud Pak for Data

4.5 Additional solution templates and resources to get started with AI on Linux

Although this chapter focuses on two examples of multi-model AI solutions, other implementation patterns are also suitable for various business use cases.

The AI on IBM Z team, in collaboration with the Open Mainframe Project at the Linux Foundation through the Ambitus project, offers solution templates for a range of AI on IBM Z use cases. These AI solution templates help you accelerate the development, deployment, and production planning of AI models and applications on IBM Z and LinuxONE.

- For a template for fraud detection that uses the AI Toolkit for IBM Z and LinuxONE, see this GitHub repository.
- For additional AI solution templates, see this GitHub repository.
- For more options, contact the AI on IBM Z team at aionz@us.ibm.com.

Concepts and design patterns from AI solution templates targeting z/OS are also applicable to Linux on IBM Z and LinuxONE environments.

Abbreviations and acronyms

AI	artificial intelligence
AML	anti-money laundering
CSV	comma-separated value
DDL	data definition language
DL	deep learning
DLC	deep learning compiler
DOU	document of understanding
DPU	data processing unit
GAN	Generative Adversarial Network
GBM	Gradient Boosting Machines
gRPC	Google Remote Procedure Call
IBM	International Business Machines Corporation
KPI	key performance indicator
KYC	Know-Your-Customer
LLM	large language model
ML	machine learning
IBM MLz	IBM Machine Learning for IBM z/OS
NLP	natural language processing
OLTP	online transaction processing
ONNX	Open Neural Network Exchange
PII	personally identifiable information
PMML	Predictive Model Markup Language
PMP	Project Management Professional
PoCs	Proofs of Concept
PwC	PricewaterhouseCoopers
RHCE	Red Hat Certified Engineer
SLA	service-level agreement
TCO	total cost of ownership
zCX	z/OS Container Extensions
zDLC	IBM Z Deep Learning Compiler
zIIP	IBM Z Integrated Information Processor

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